

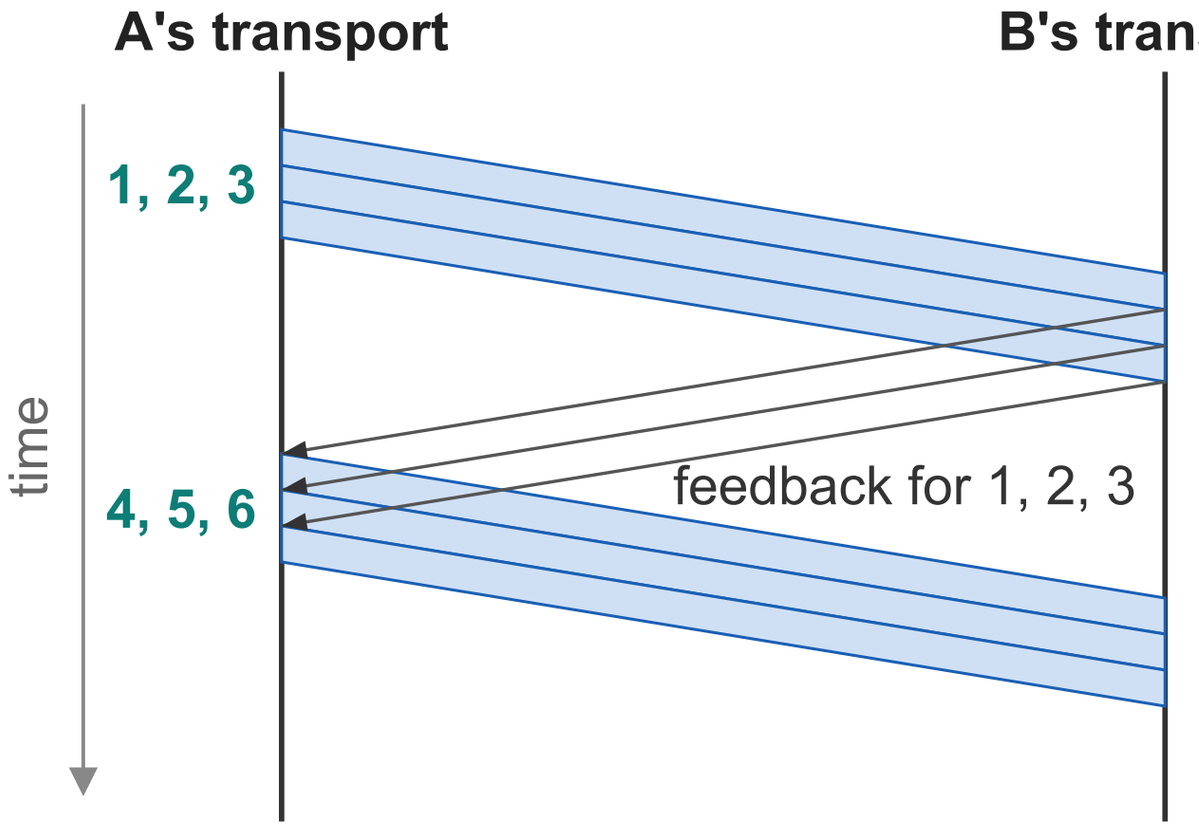
UNIVERSITY OF WATERLOO

Congestion control - Part I

CS 456/656, Fall 2026

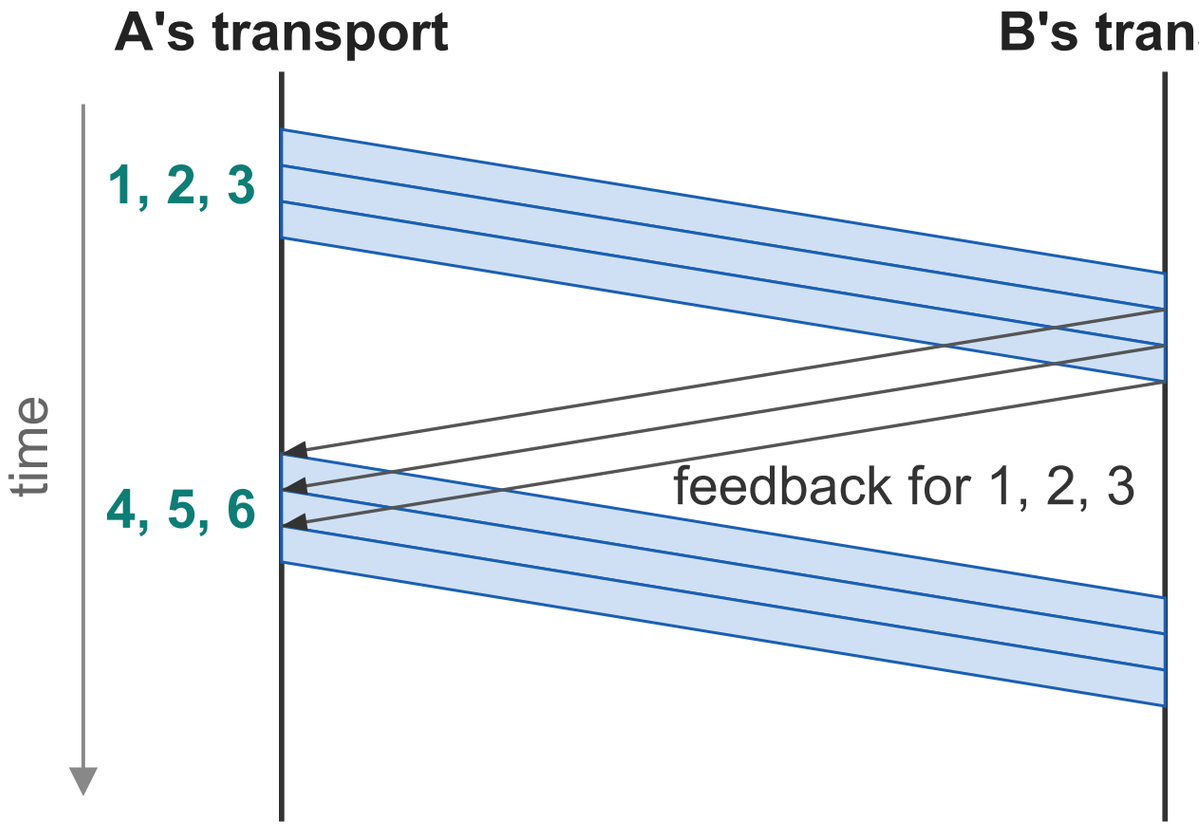
Mina Tahmasbi Arashloo

Recap: a batch of three packets at a time



A sends 3 packets back to back,
without waiting for feedback
in between.

Recap: a batch of three packets at a time



A sends 3 packets back to back, without waiting for feedback in between.

3 packets every round trip:
3 times the utilization

0.024 / (30 + 0.024) = 0.0008

Recap: what is a good batch size?

- **Too small:** we are underutilizing the network, and it takes longer to get the data delivered
- **Too big:**
  - The network is shared. If everyone sends everything they have at full speed, it will create congestion that can result in increasing drops and delays
  - The sender may “overwhelm” the receiver

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✓ **flow control with a sliding window**

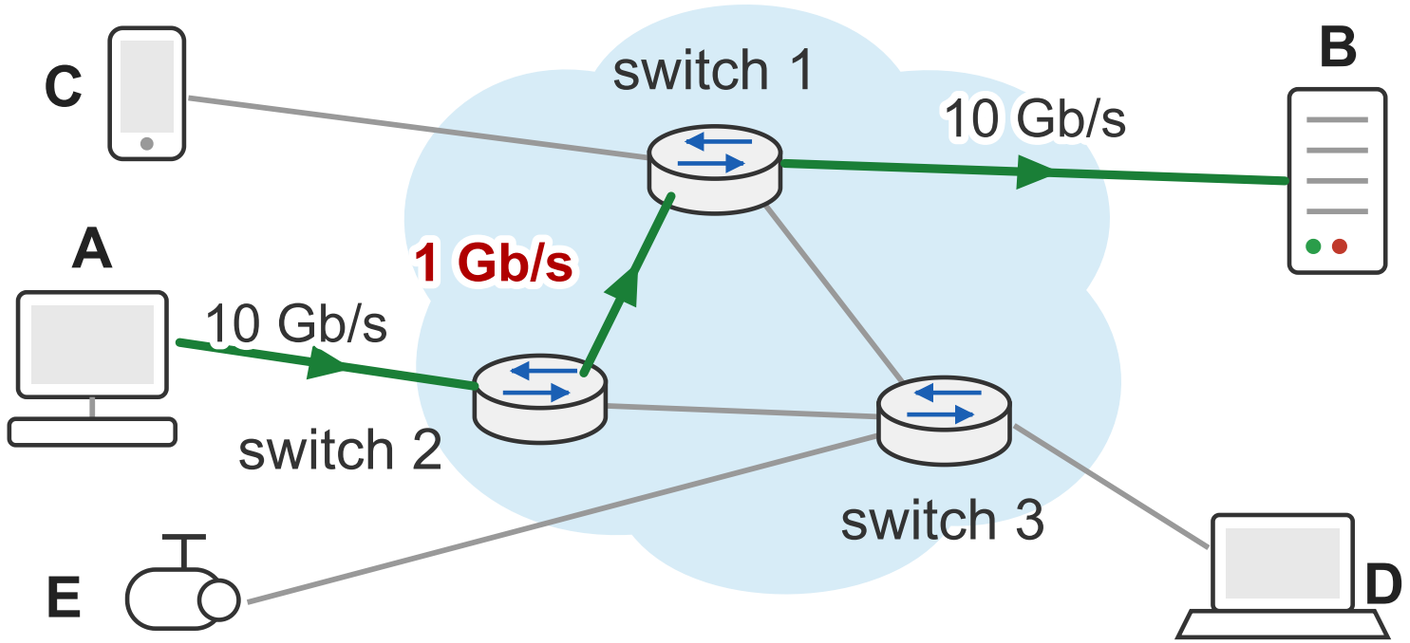
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Part 2

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✓ flow control with a sliding window

How fast should the sender send?



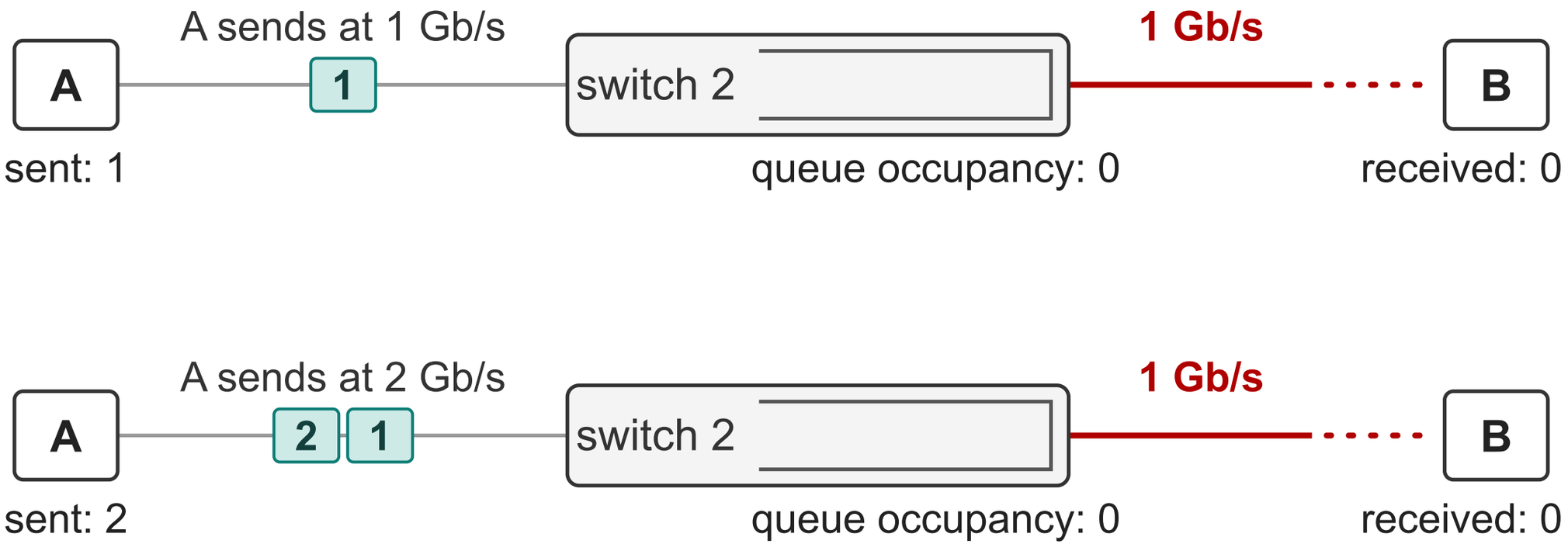
For intuition, we first talk about **sending rate**, not batch size (rate ≈ batch size / RTT).

Consider a sender and a receiver:

- The sender knows exactly what path its traffic will take to the destination, and the properties (rate, propagation delay, etc.) of each link
- The sender always has more data to send

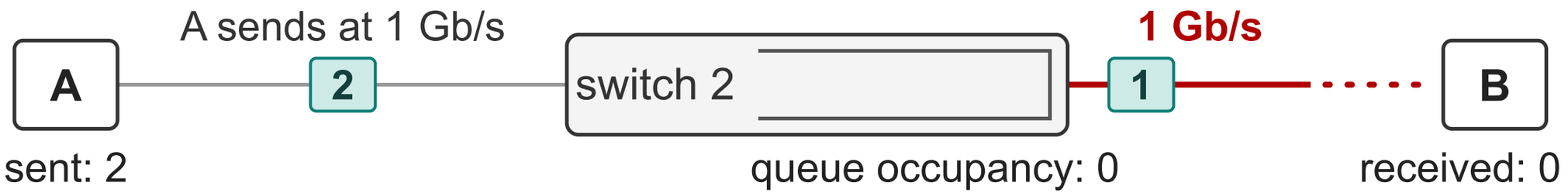
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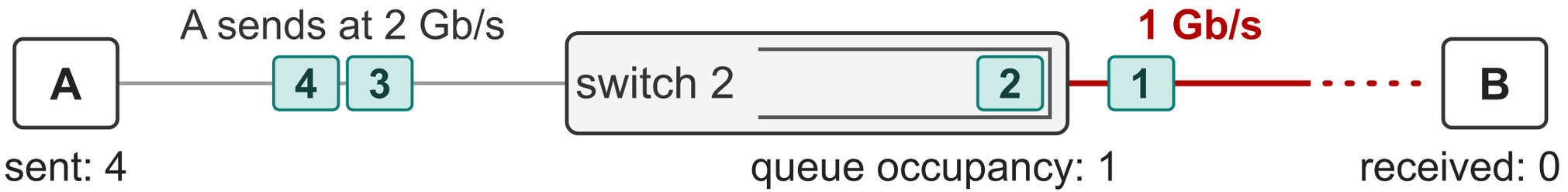
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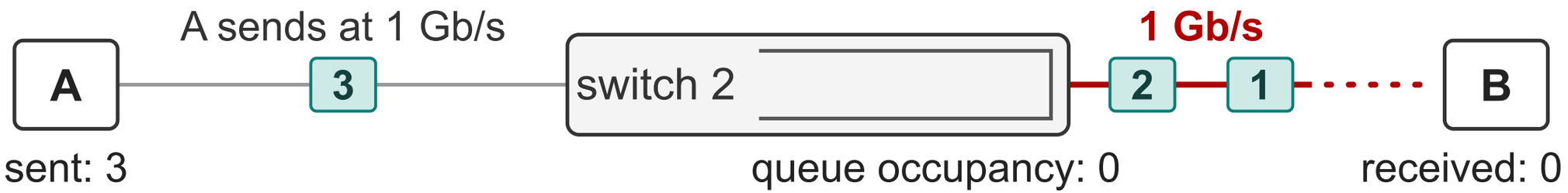
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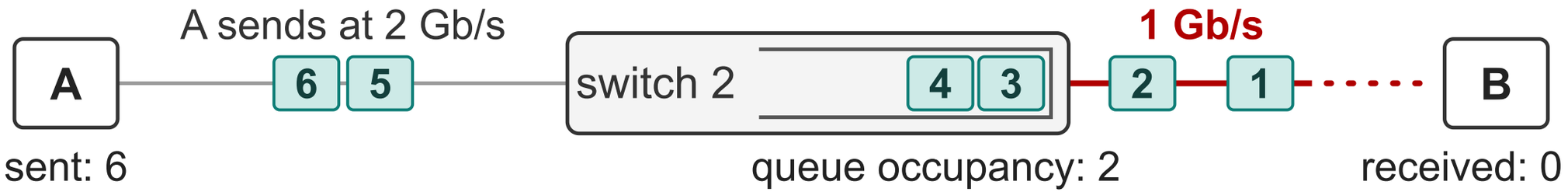




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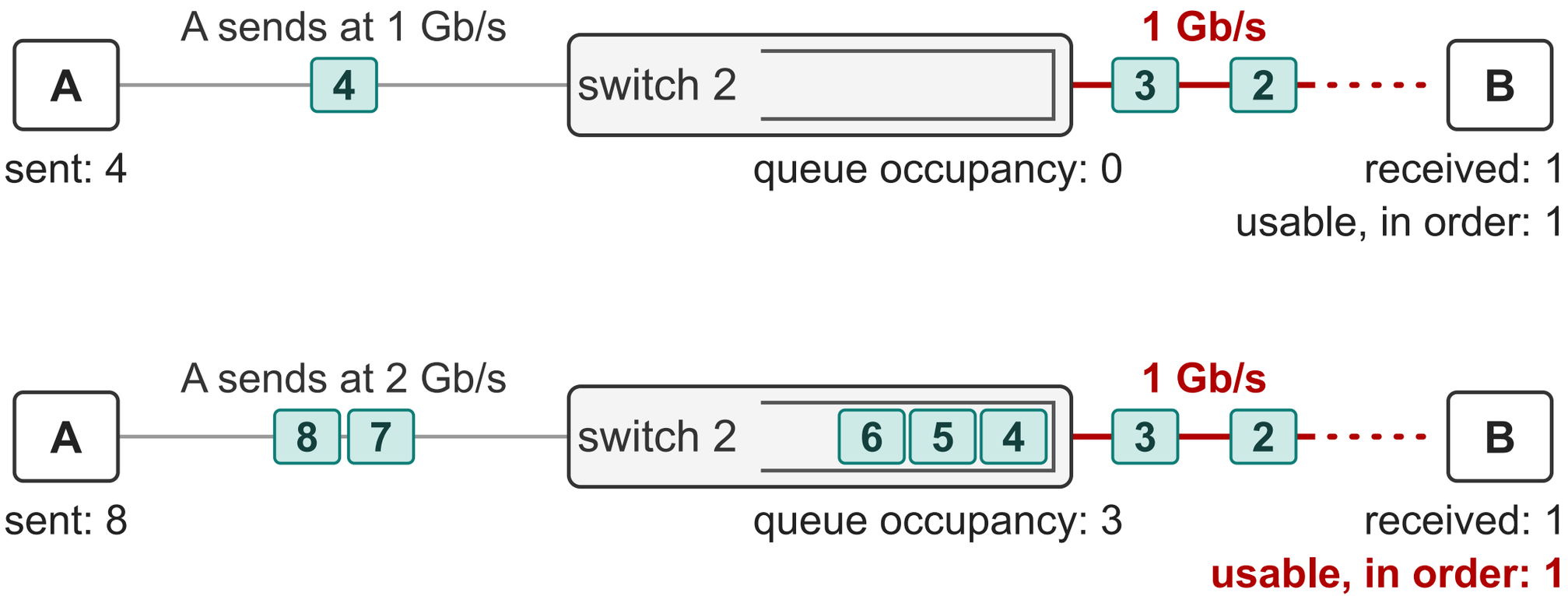
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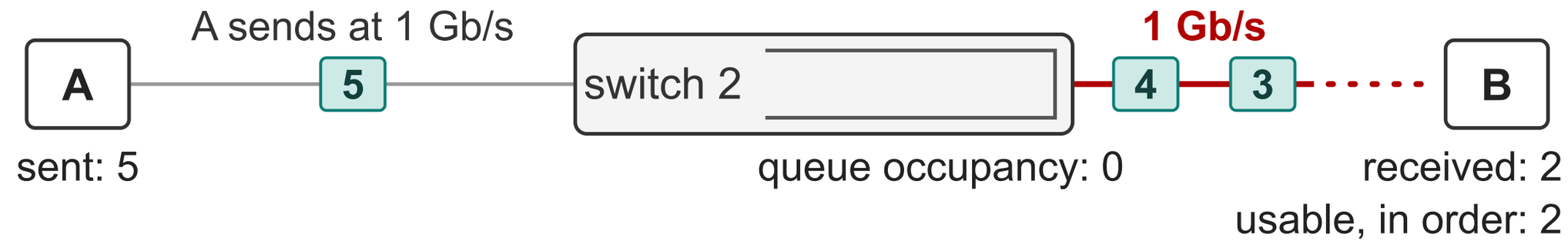
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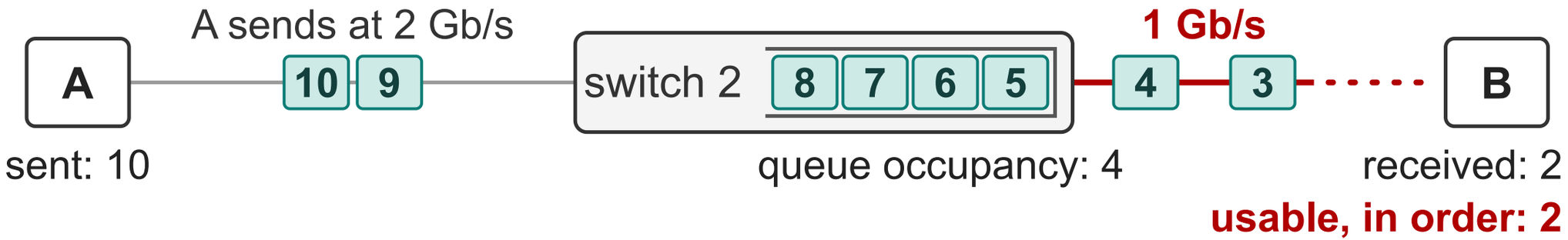
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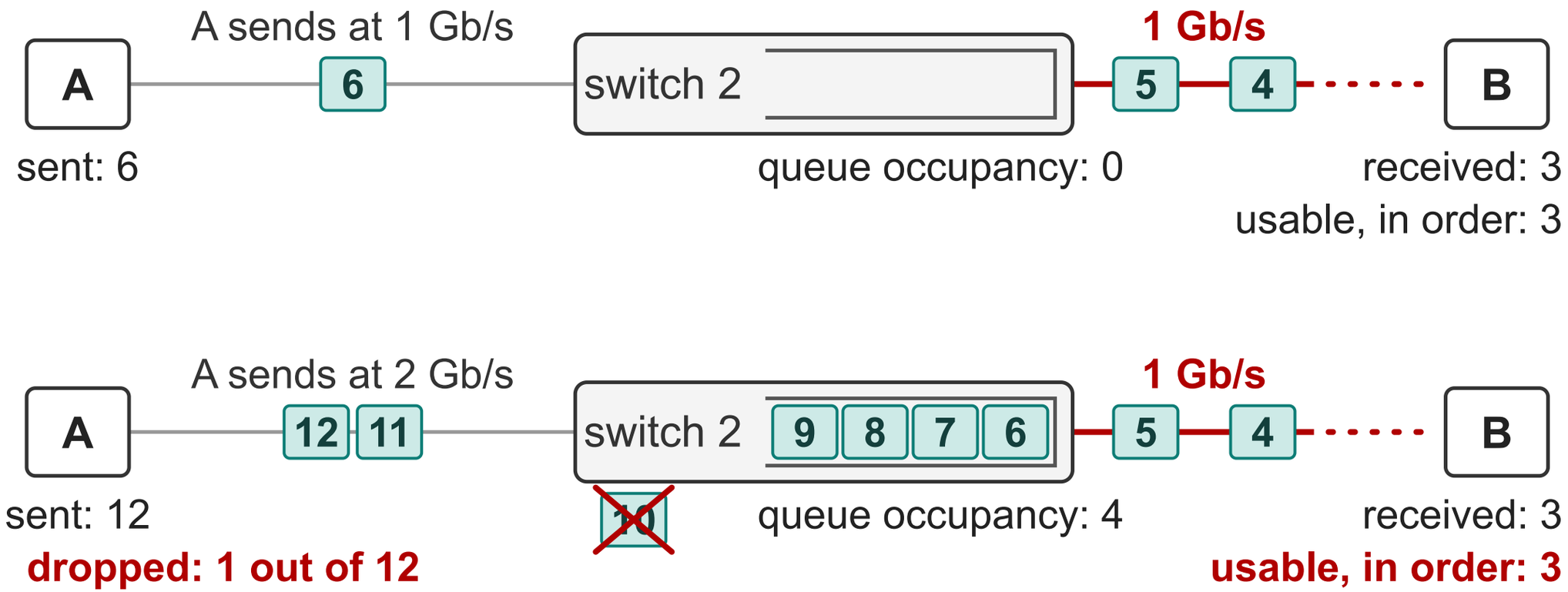
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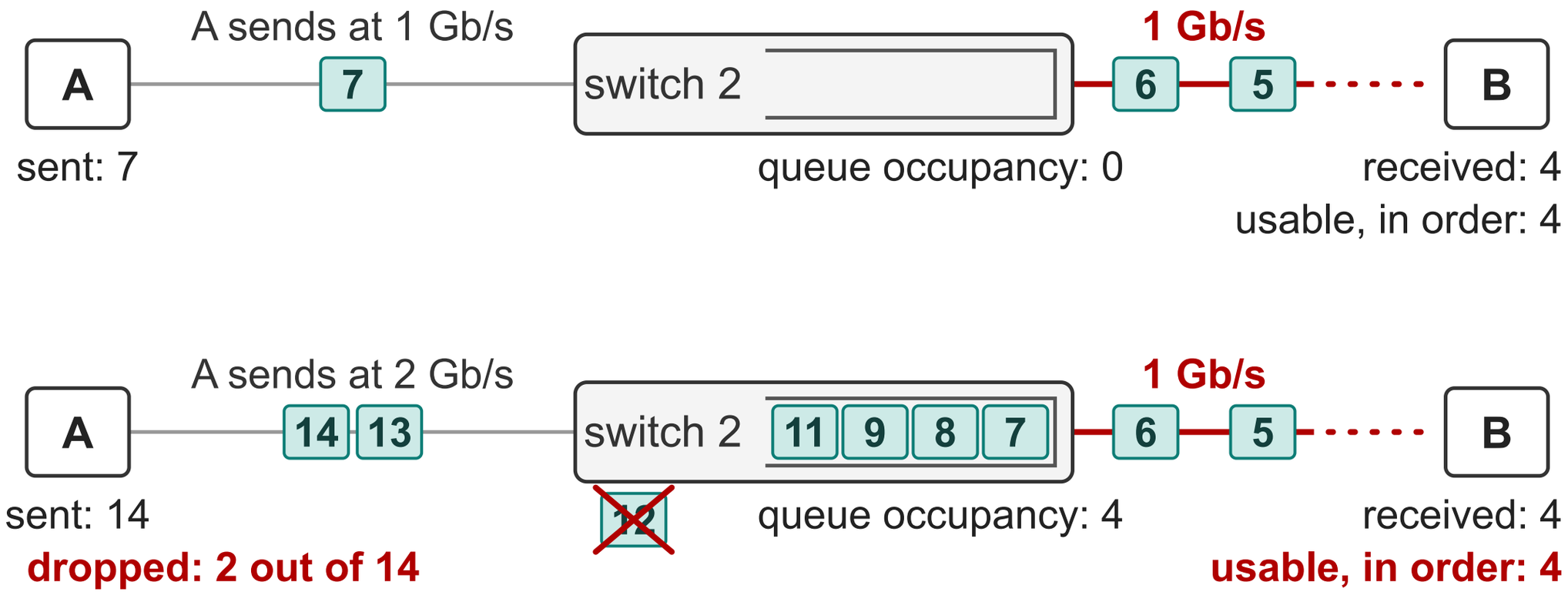
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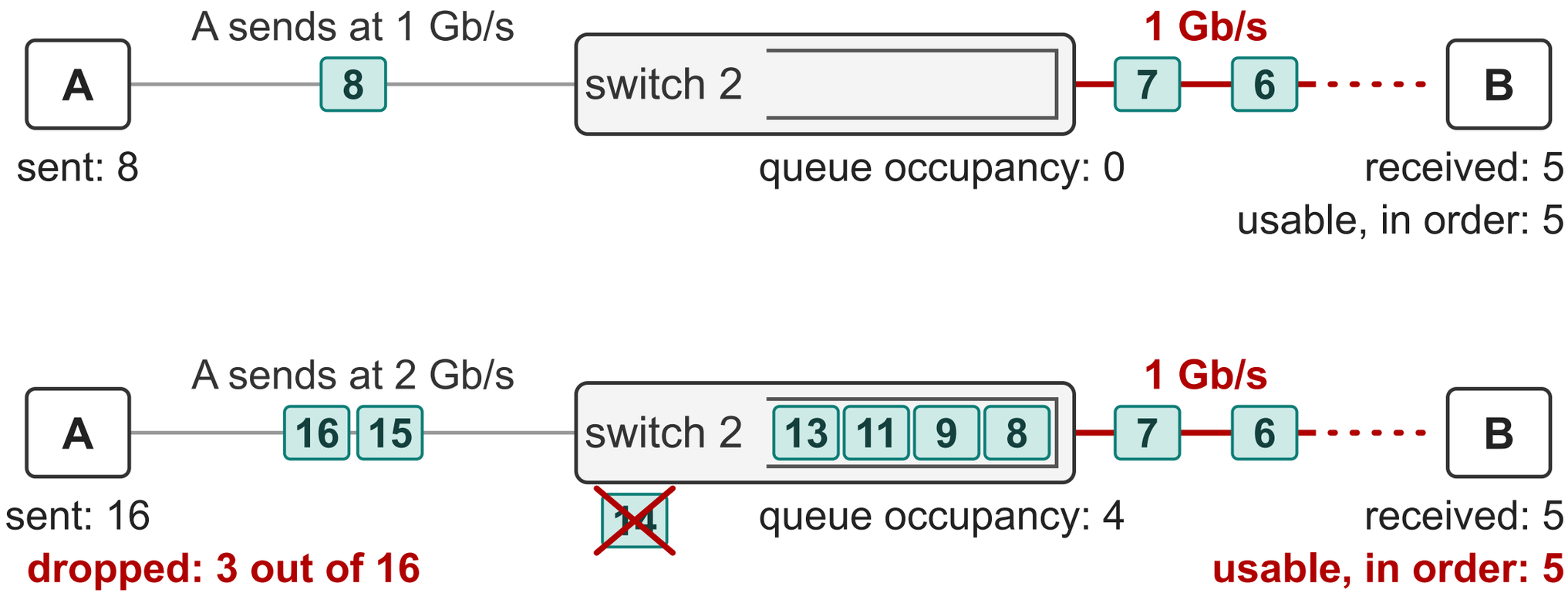
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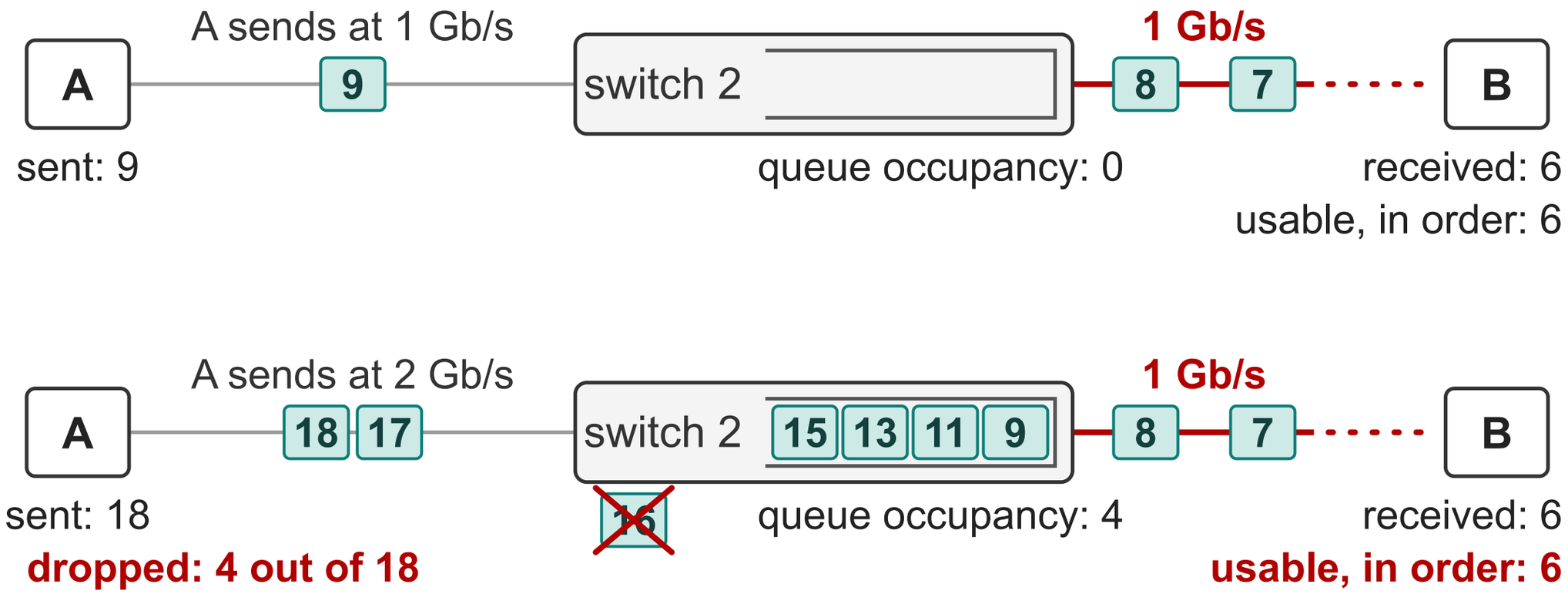
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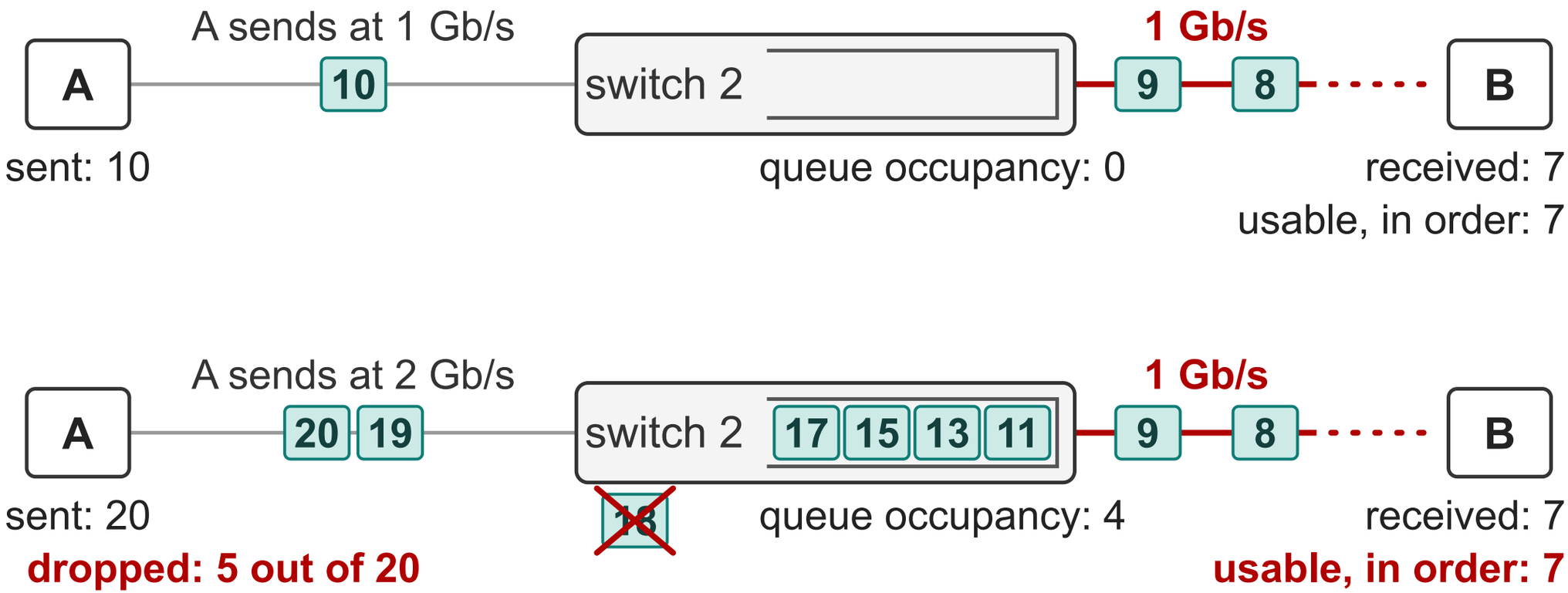
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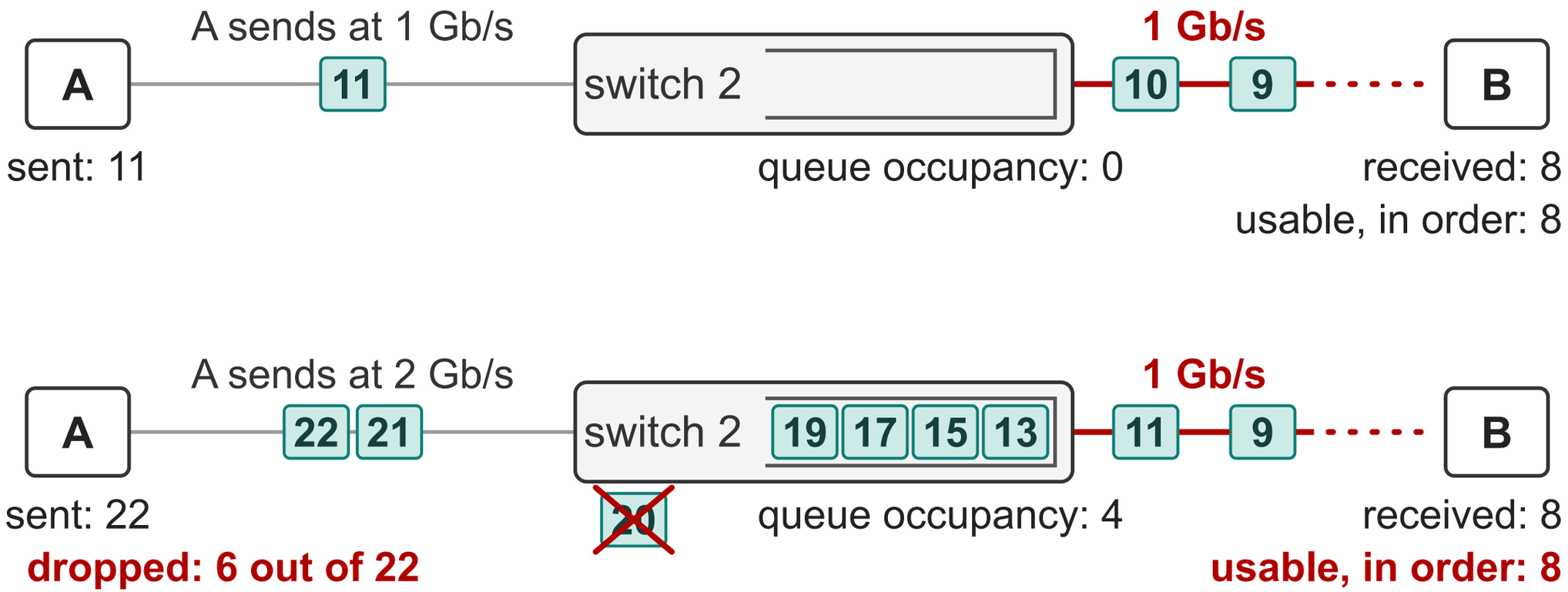
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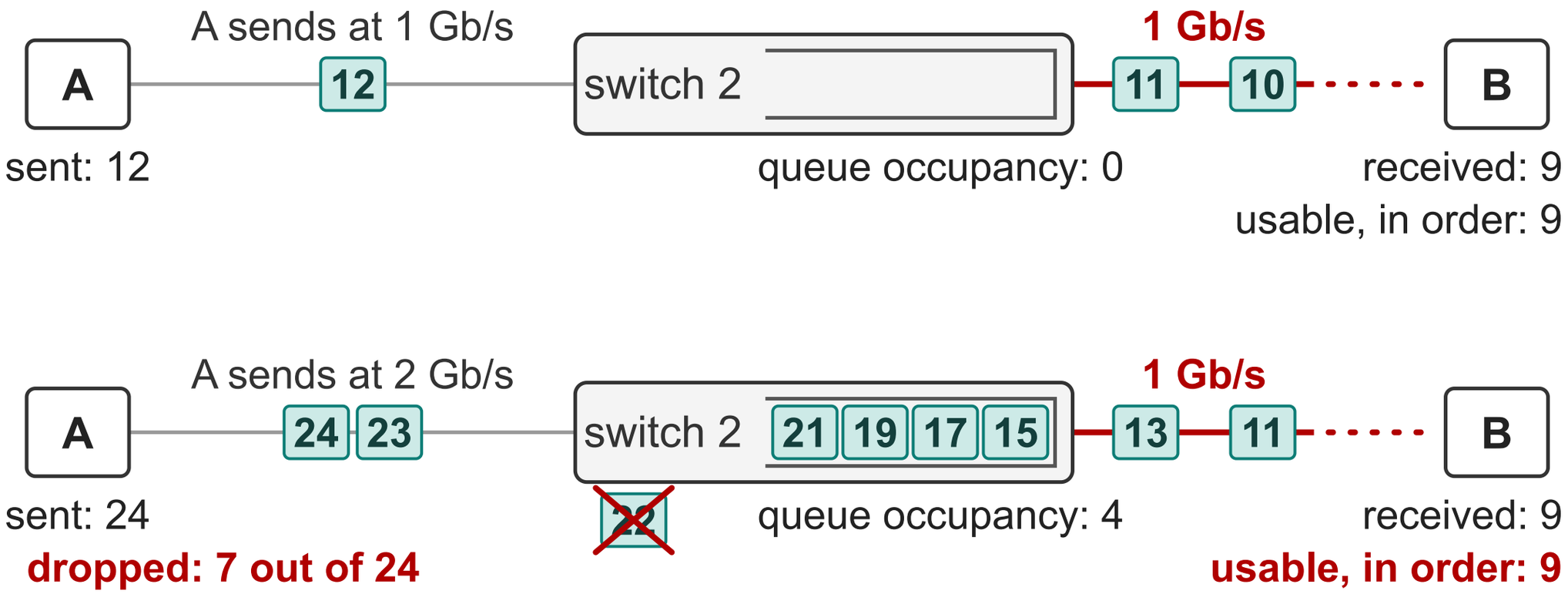
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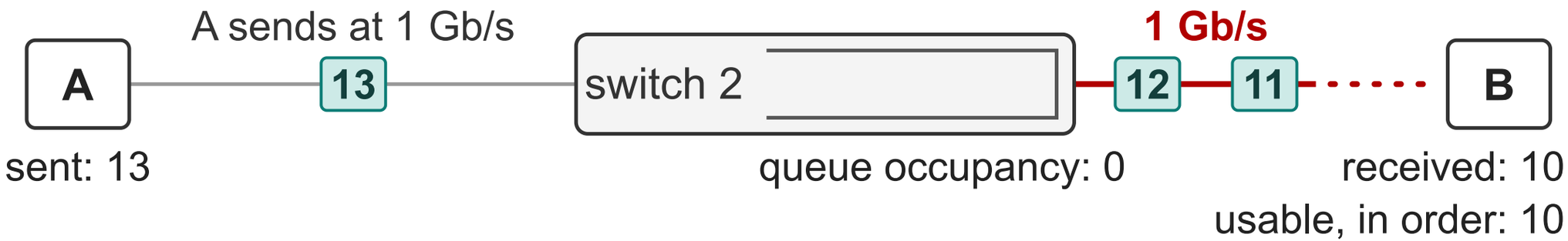
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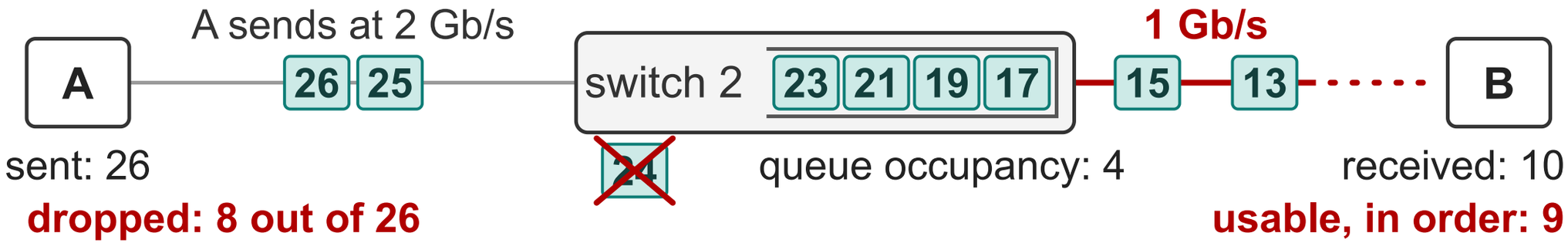
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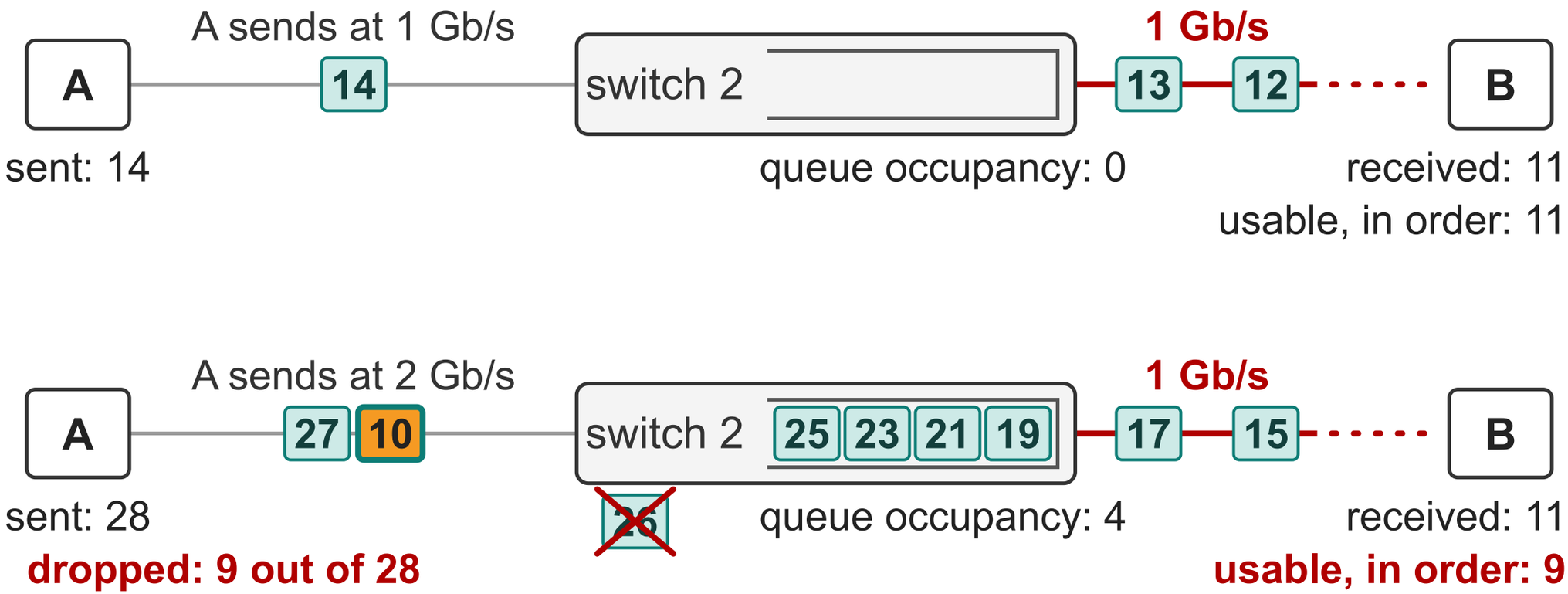
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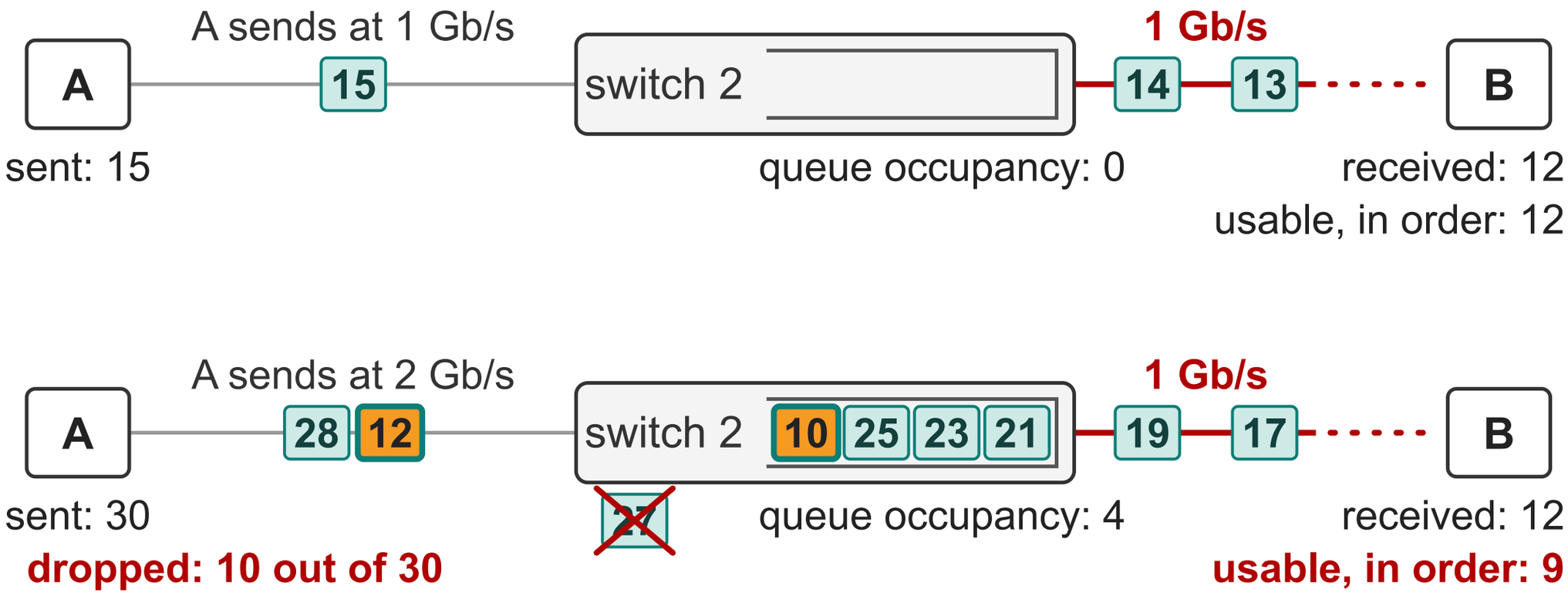
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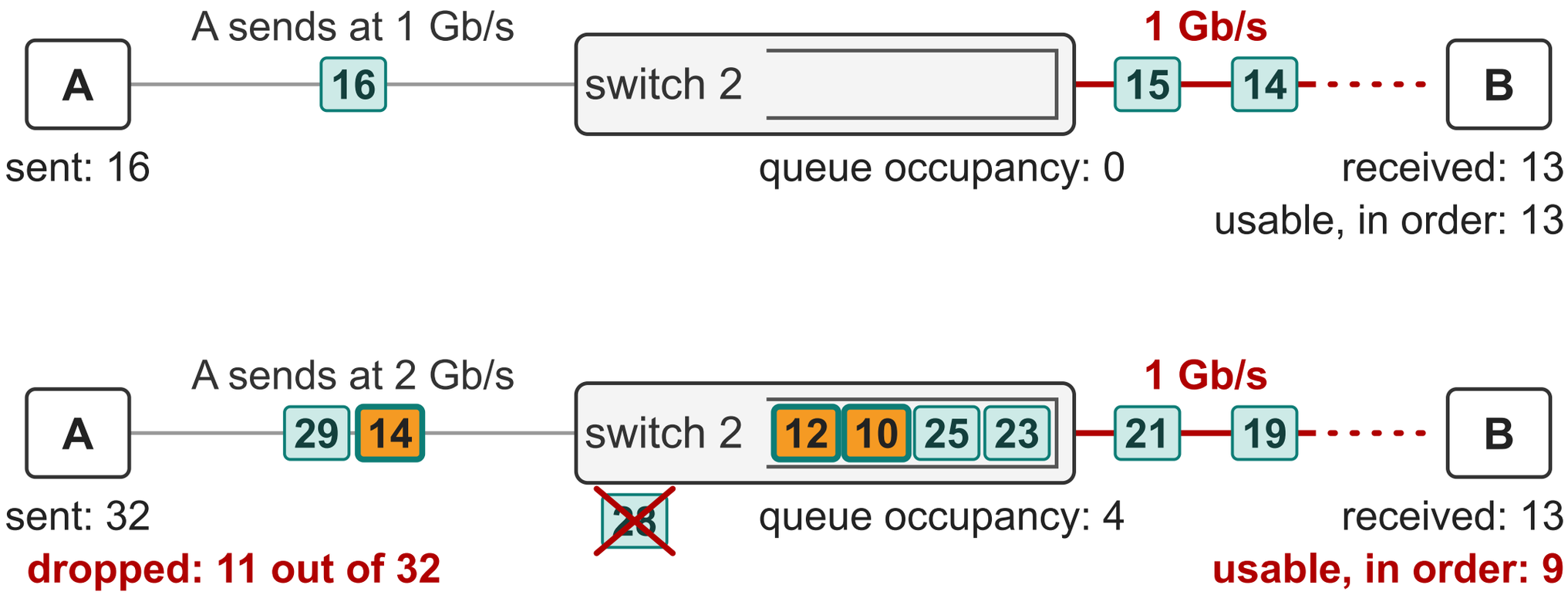
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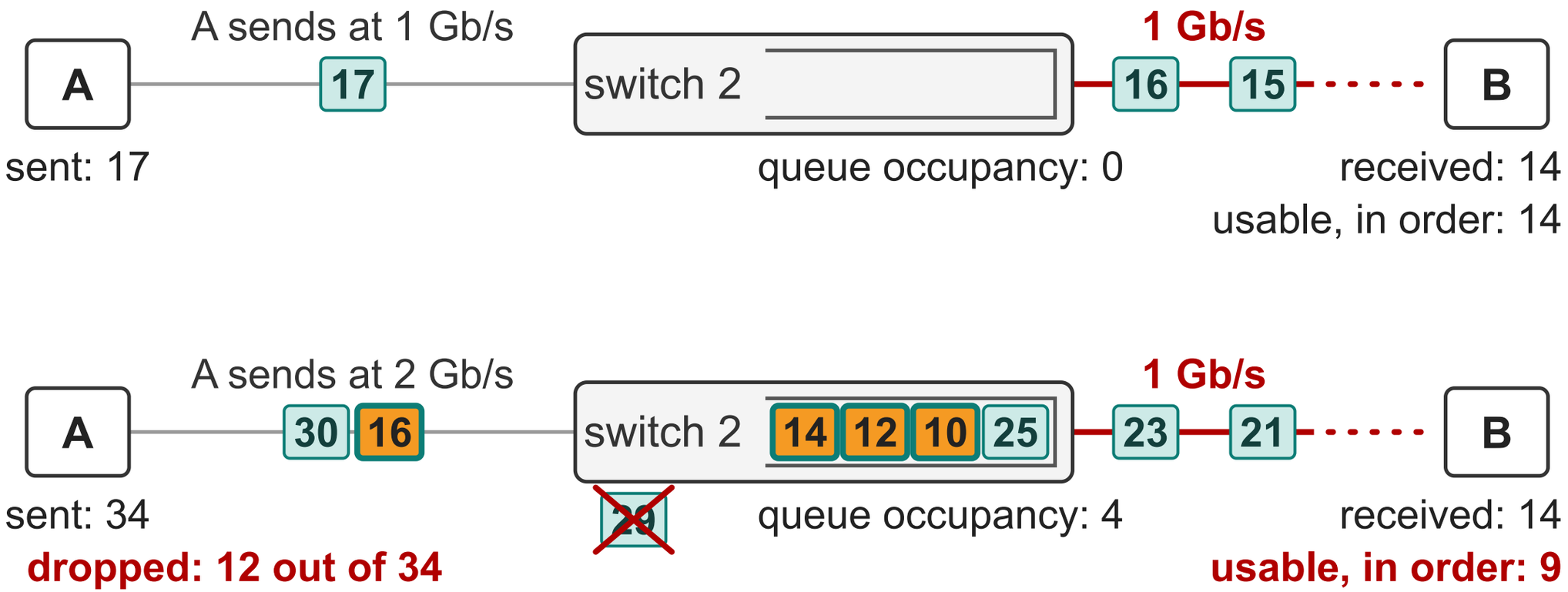
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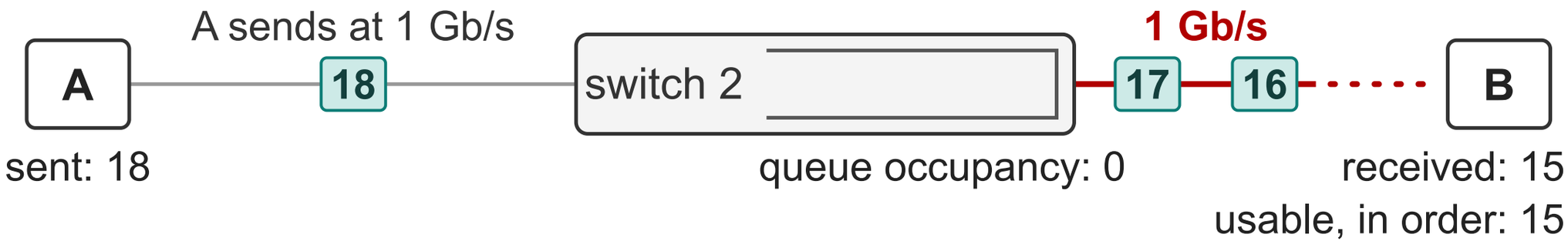
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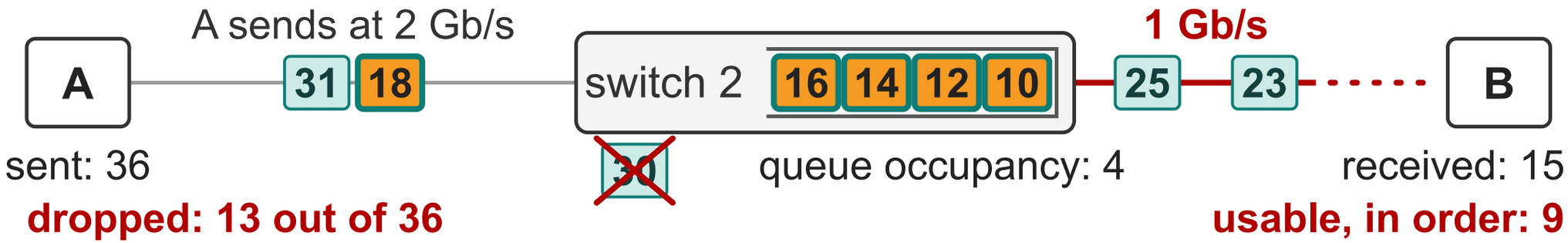
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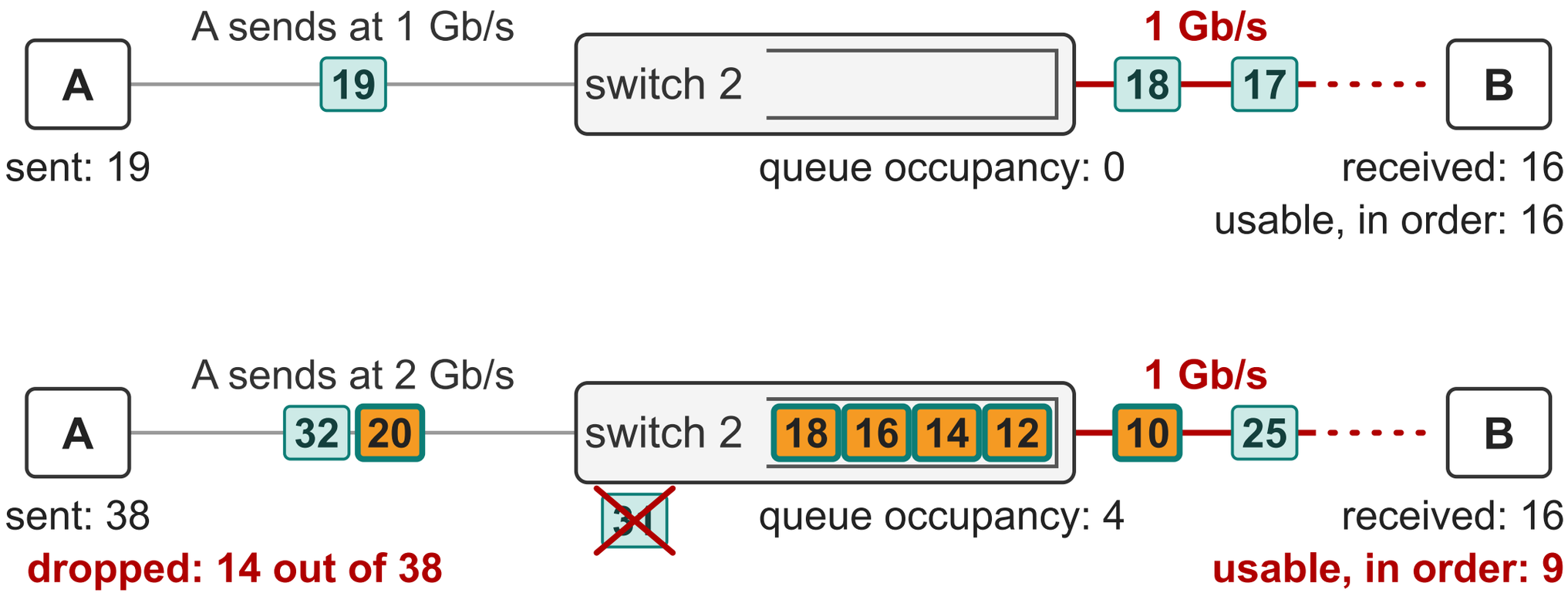
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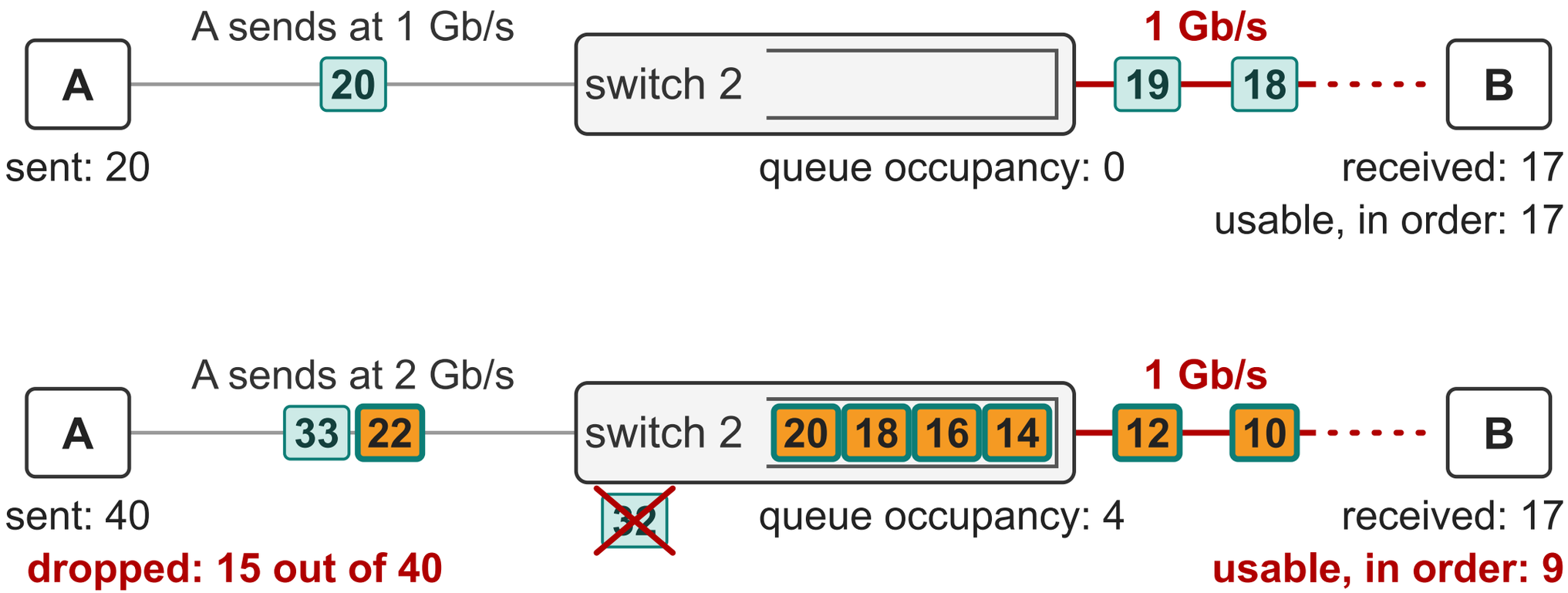
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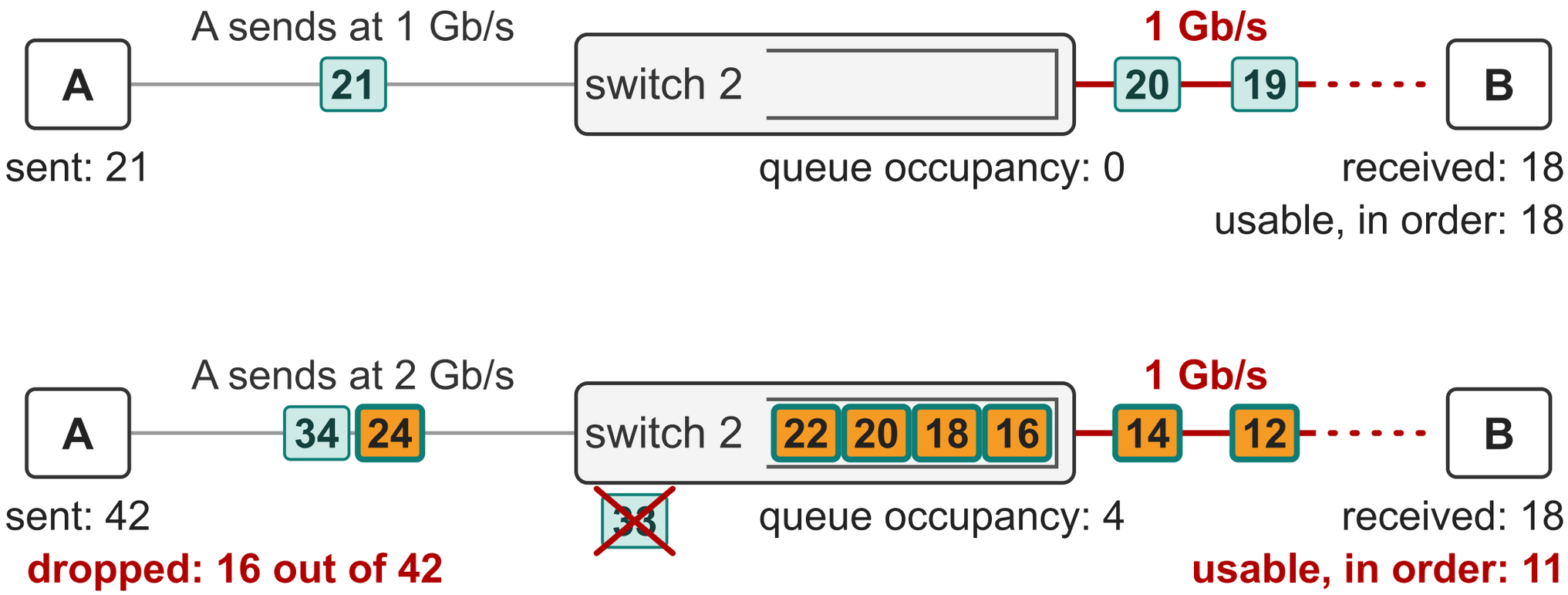
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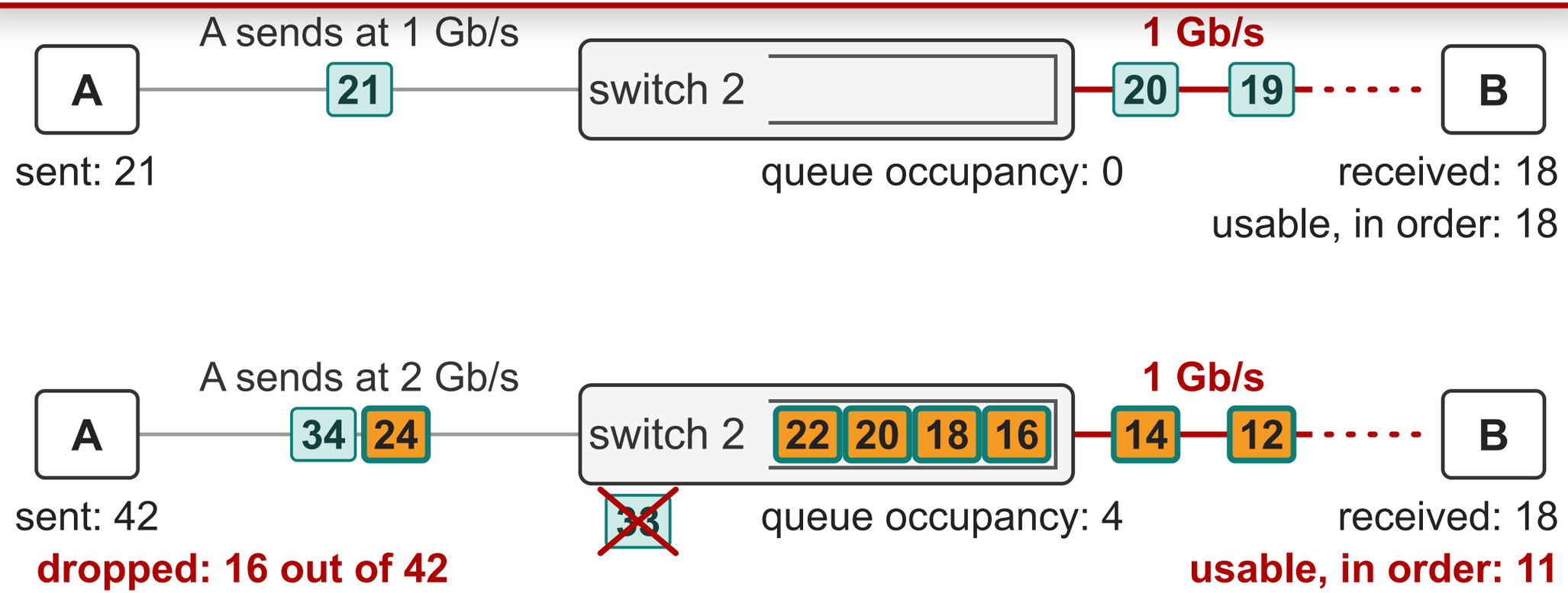
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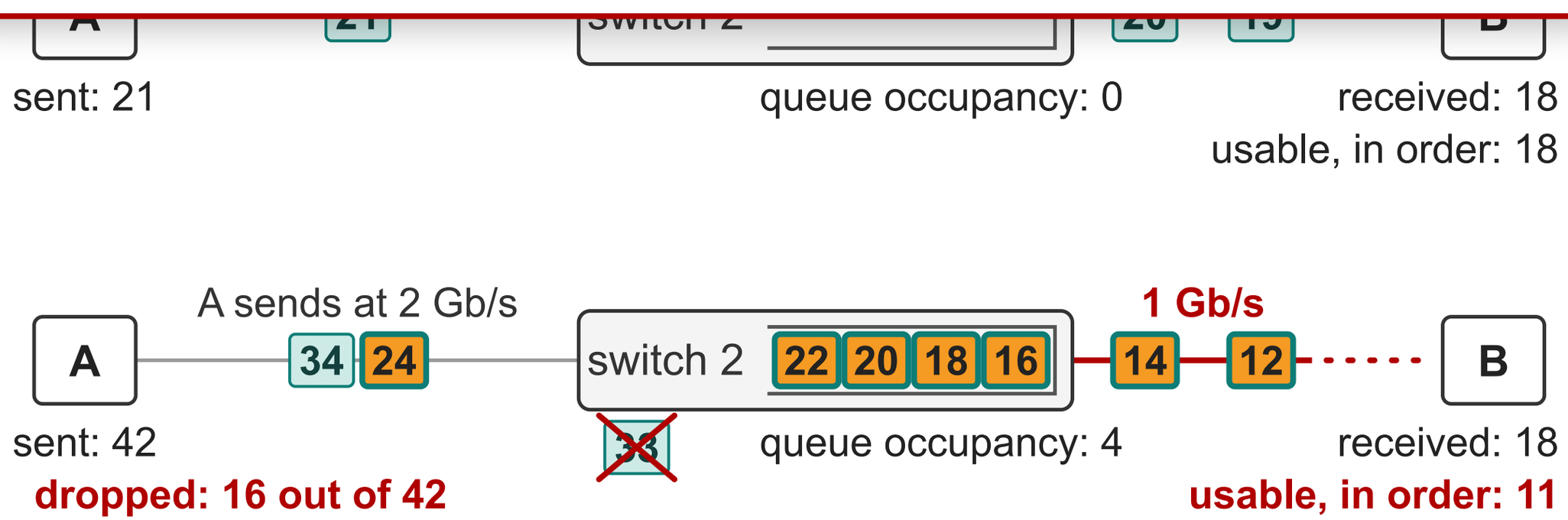


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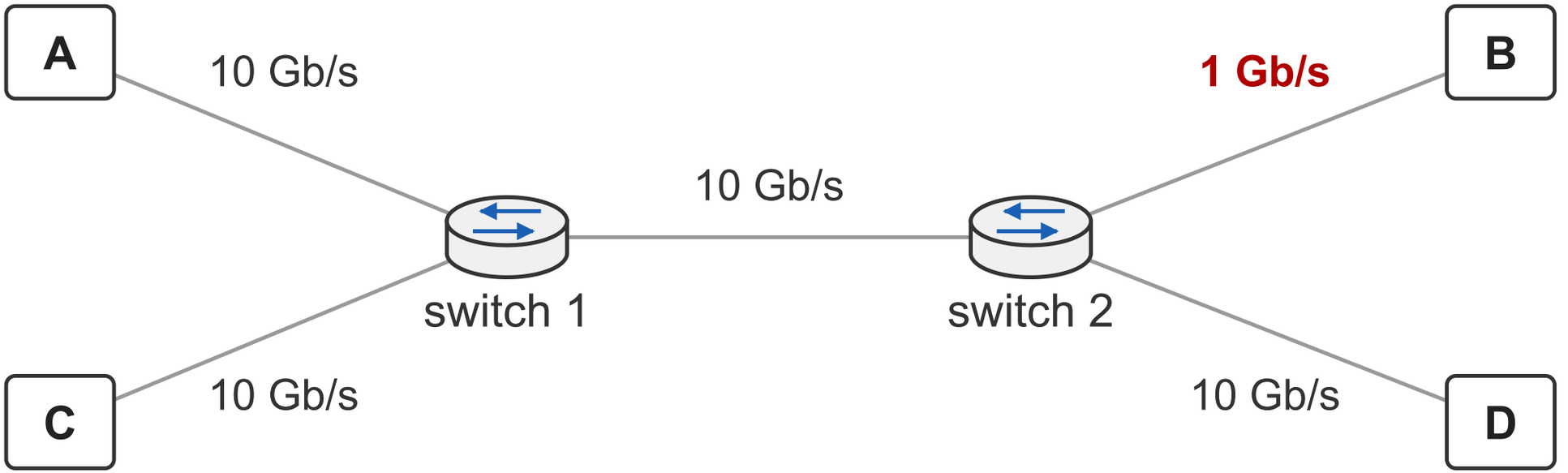
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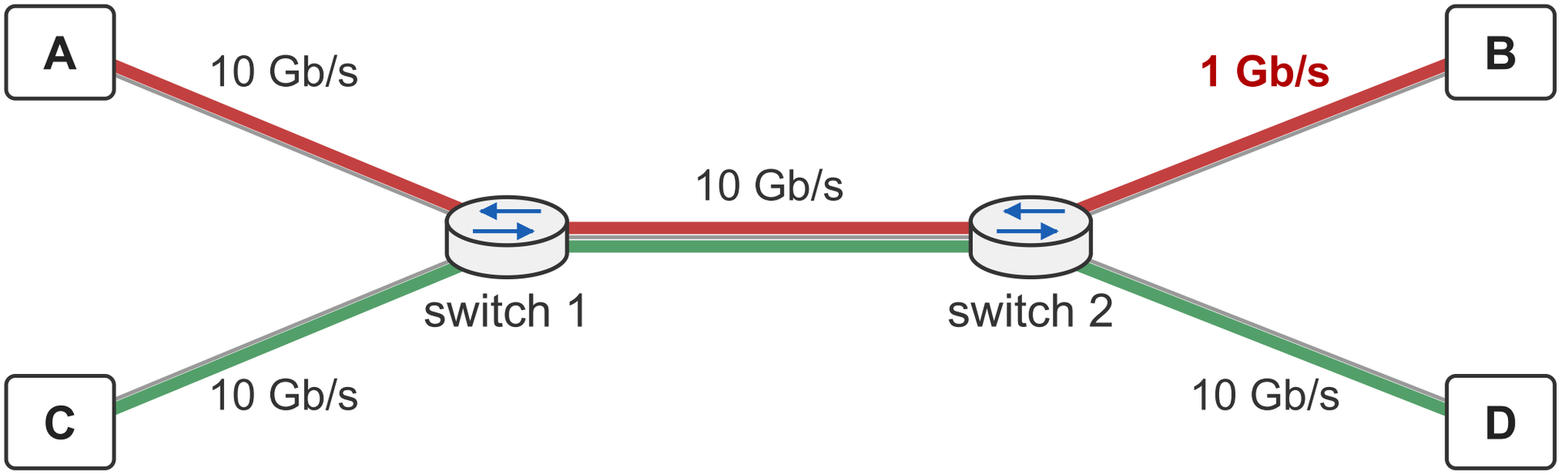
It can have severe consequences on a shared network.



Losses downstream waste upstream work

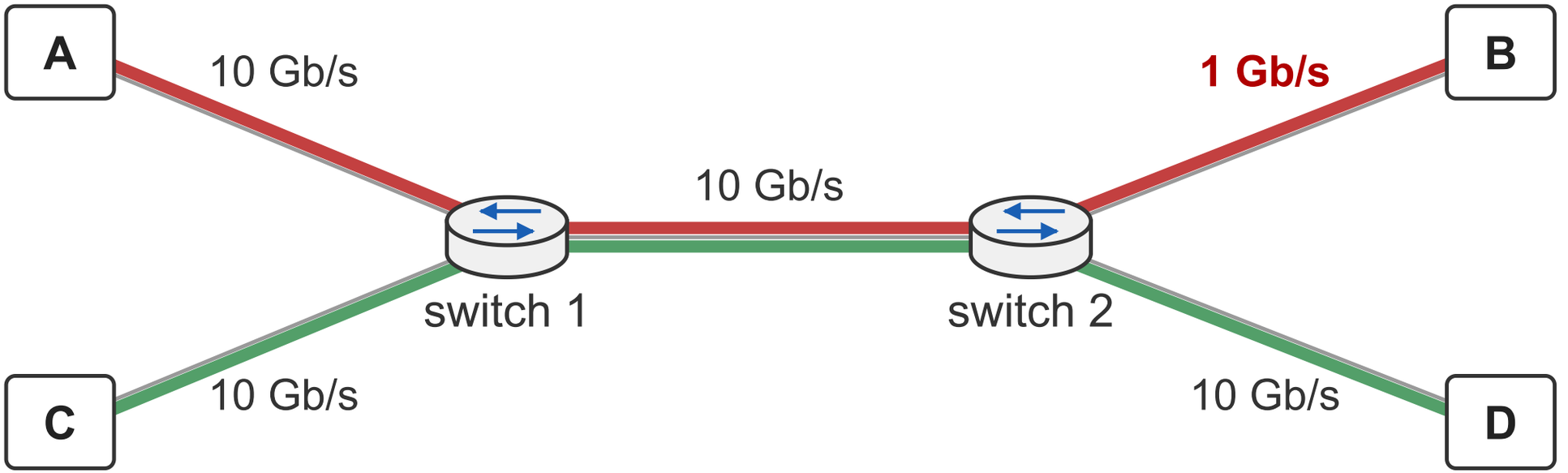


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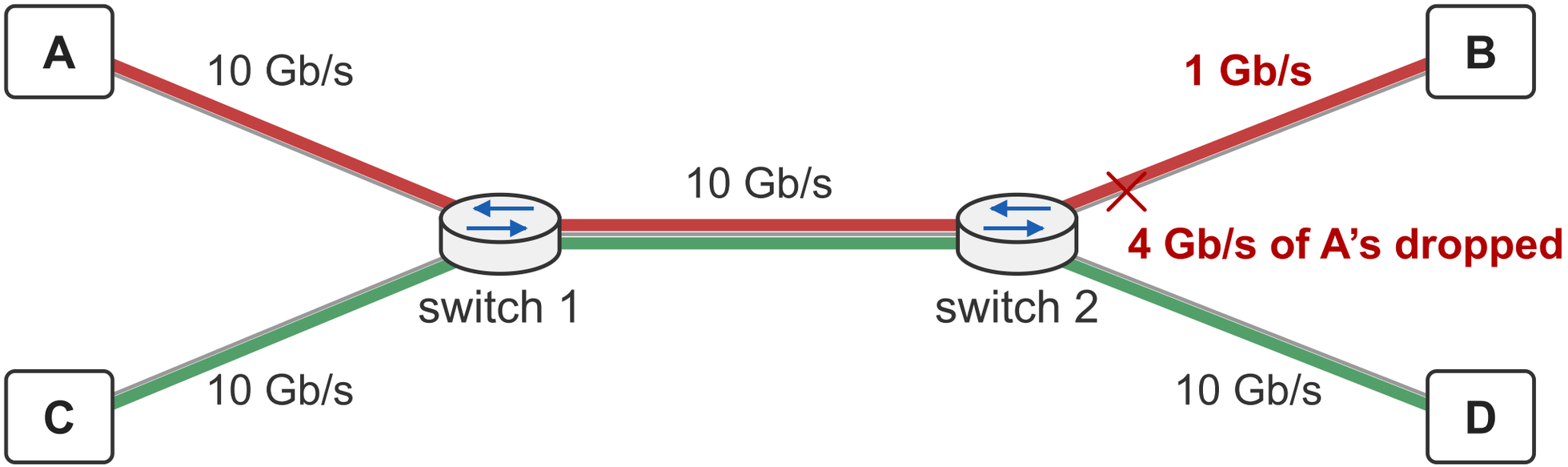
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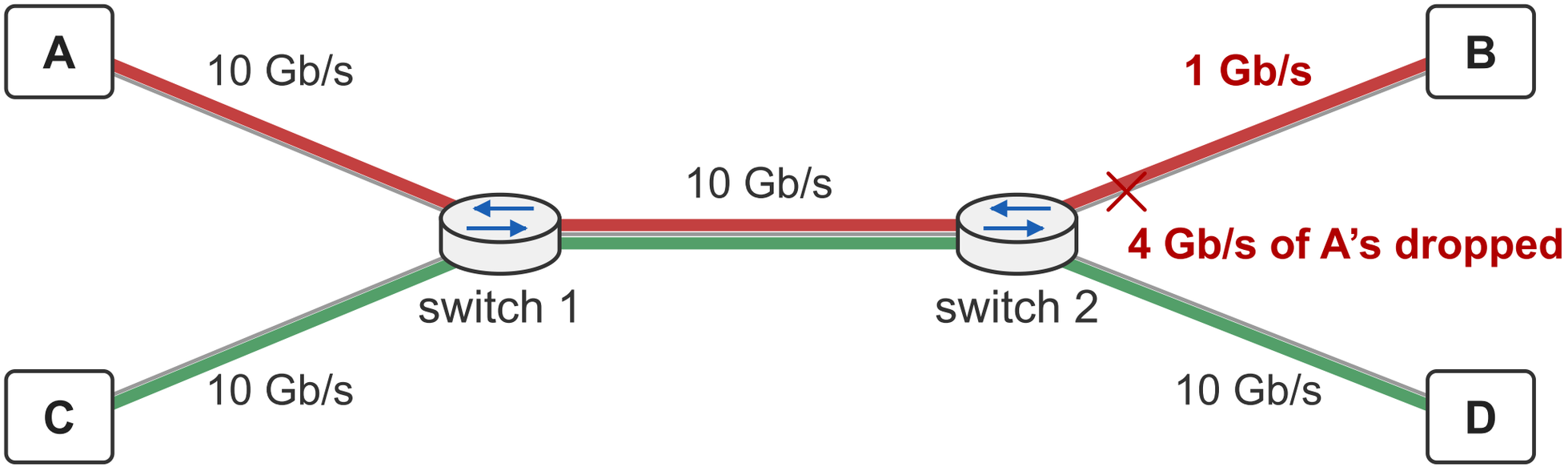
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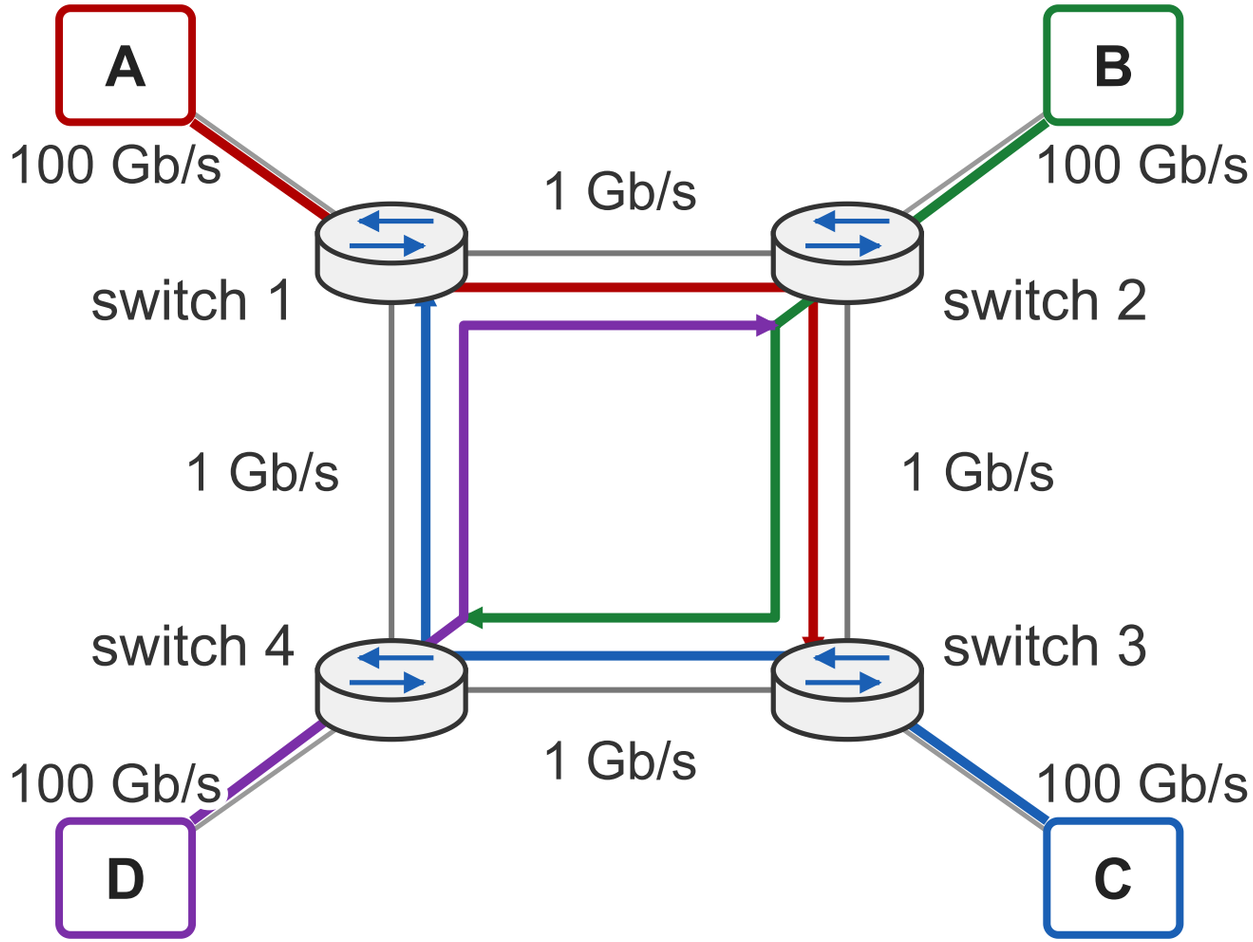
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B gets 1 Gb/s and D gets 5 Gb/s: 6 Gb/s total
Had A sent 1 Gb/s from the beginning, the total could have been 1 + 9 = 10 Gb/s

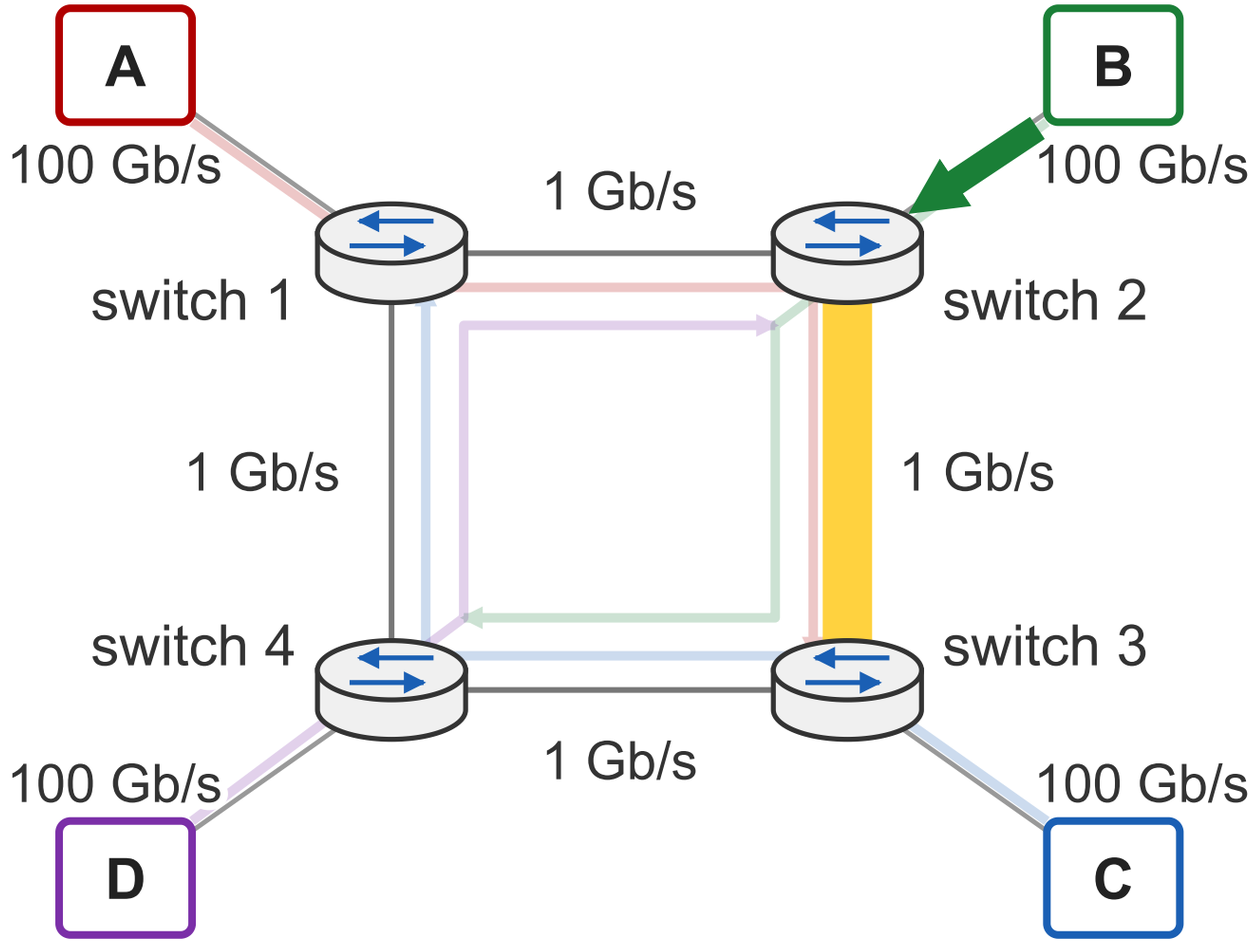
Congestion collapse

Each host sends to the opposite corner (A → C, B → D, C → A, D → B), as fast as it can.



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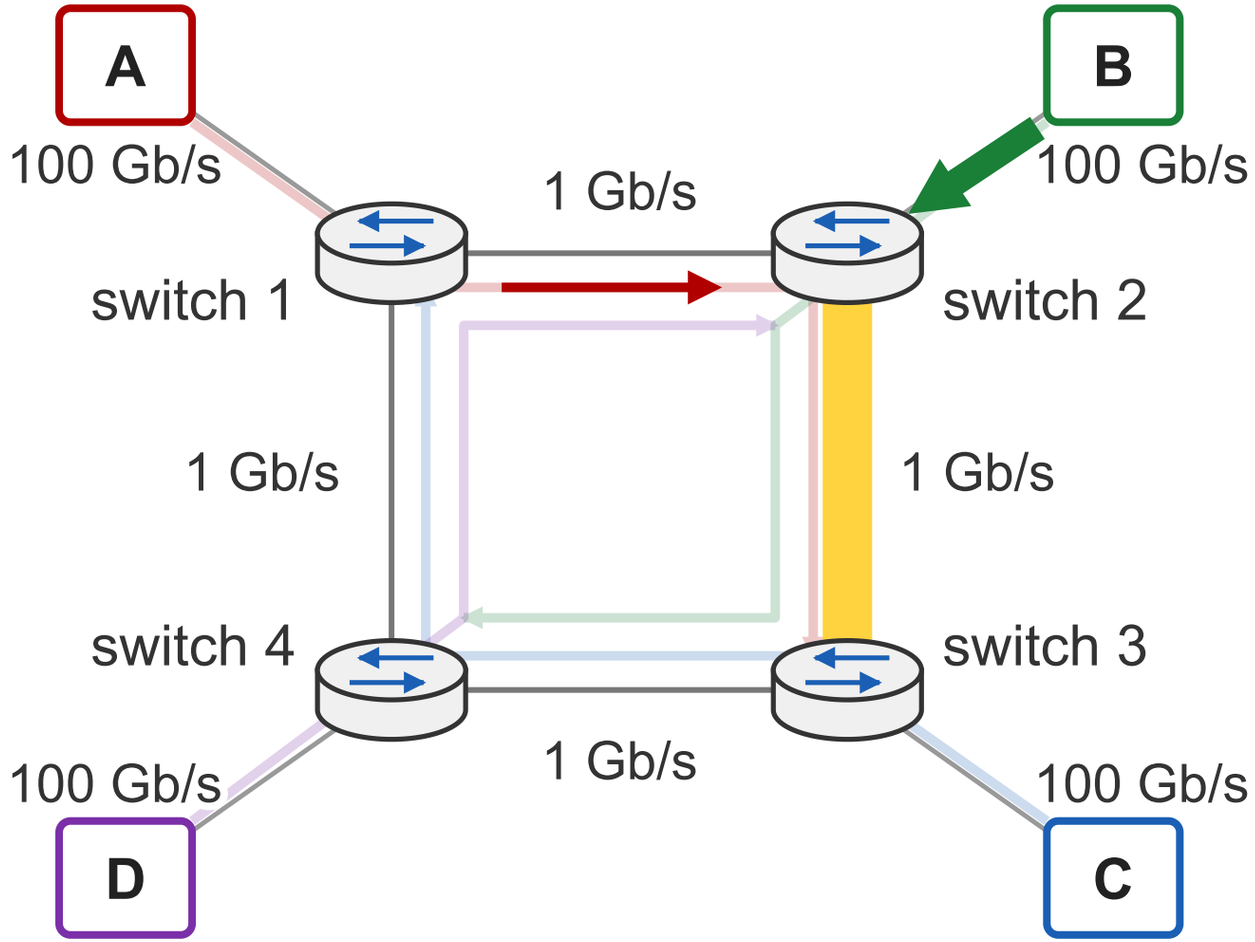


Consider switch 2's queue for the link to switch 3:

- B is sending 100 Gb/s into the queue

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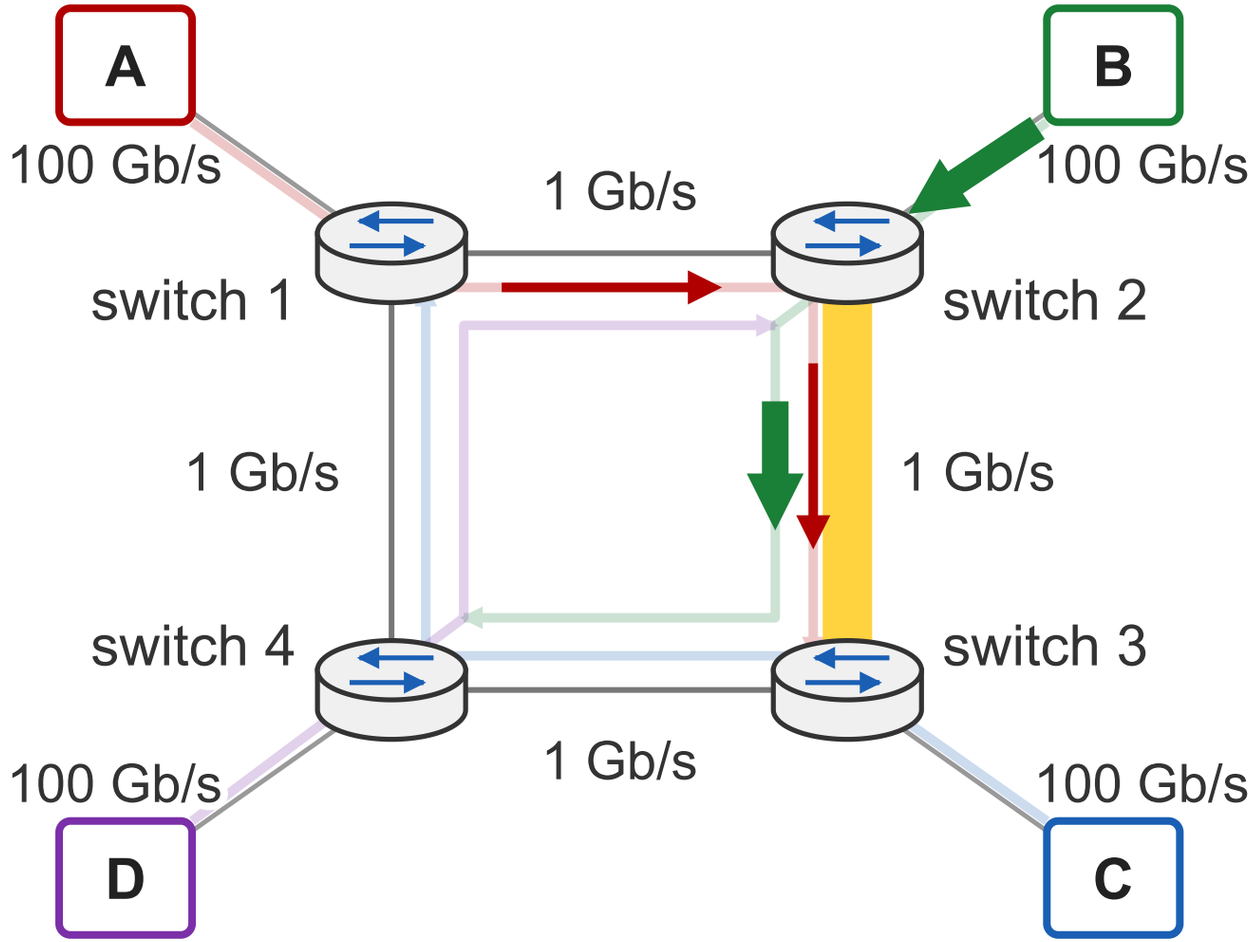


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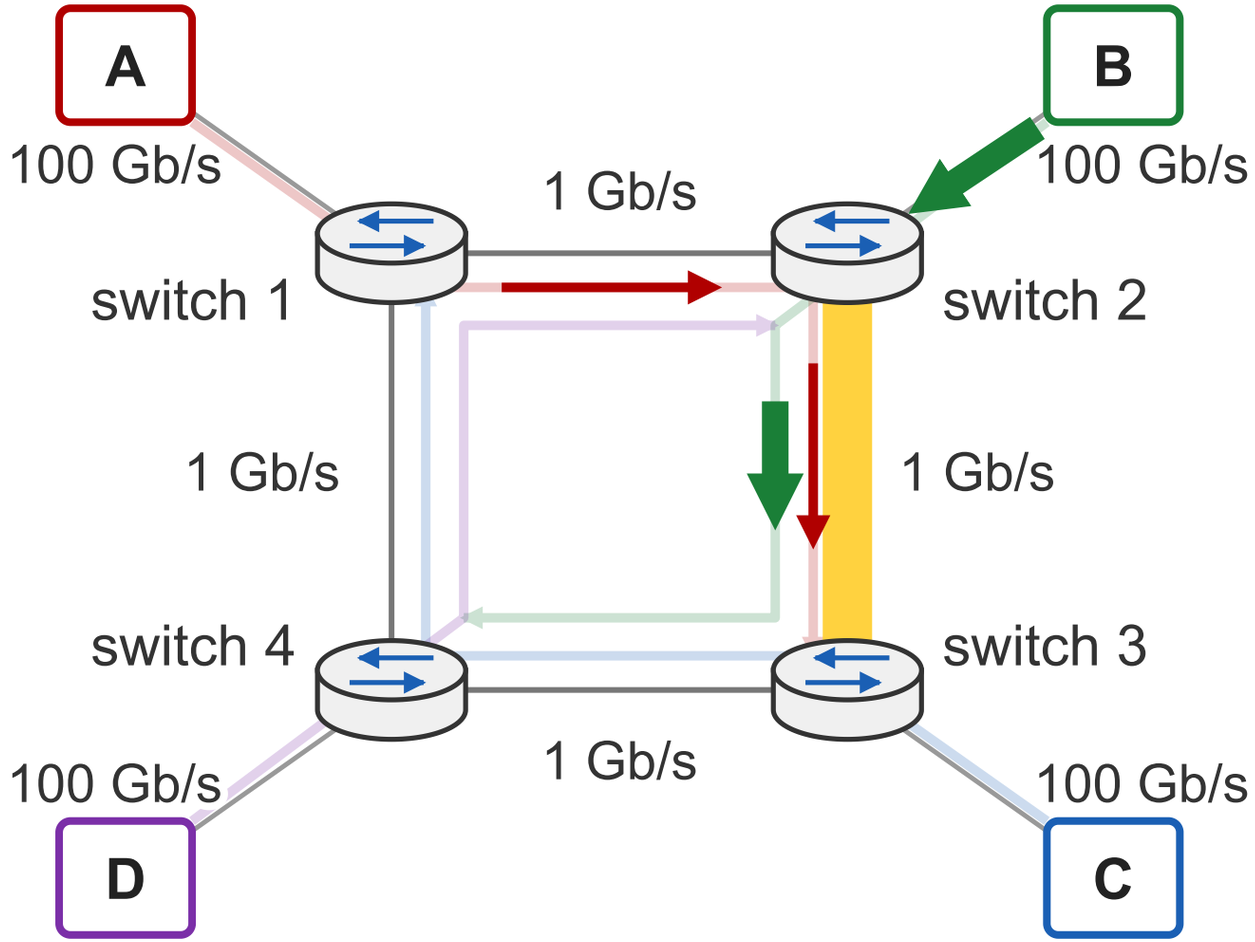


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- B is sending 100 Gb/s into the queue
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- A's share = $X / (X + 100)$
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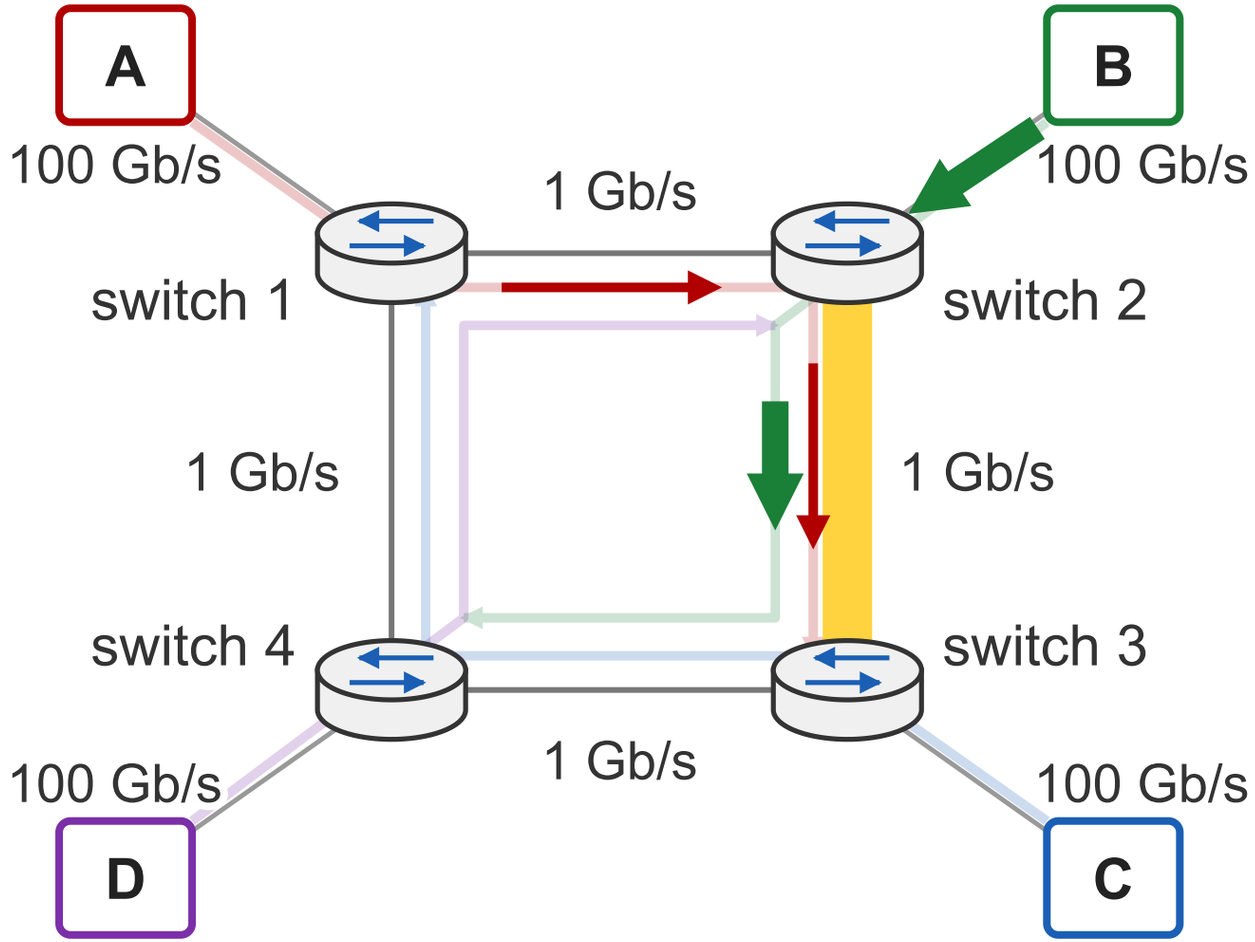


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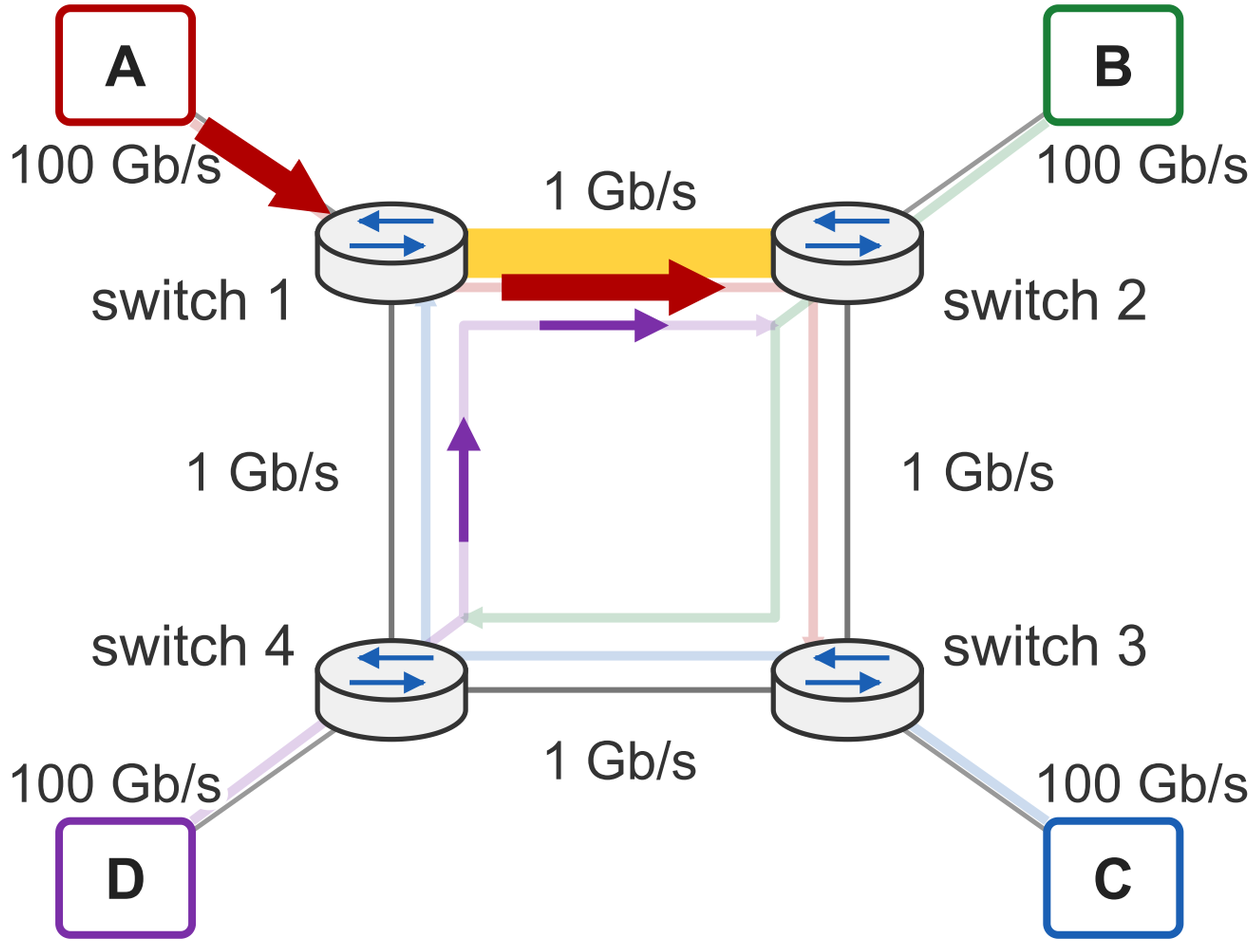
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What is X?

Congestion collapse

Each host sends to the opposite corner (A → C, B → D, C → A, D → B), as fast as it can.

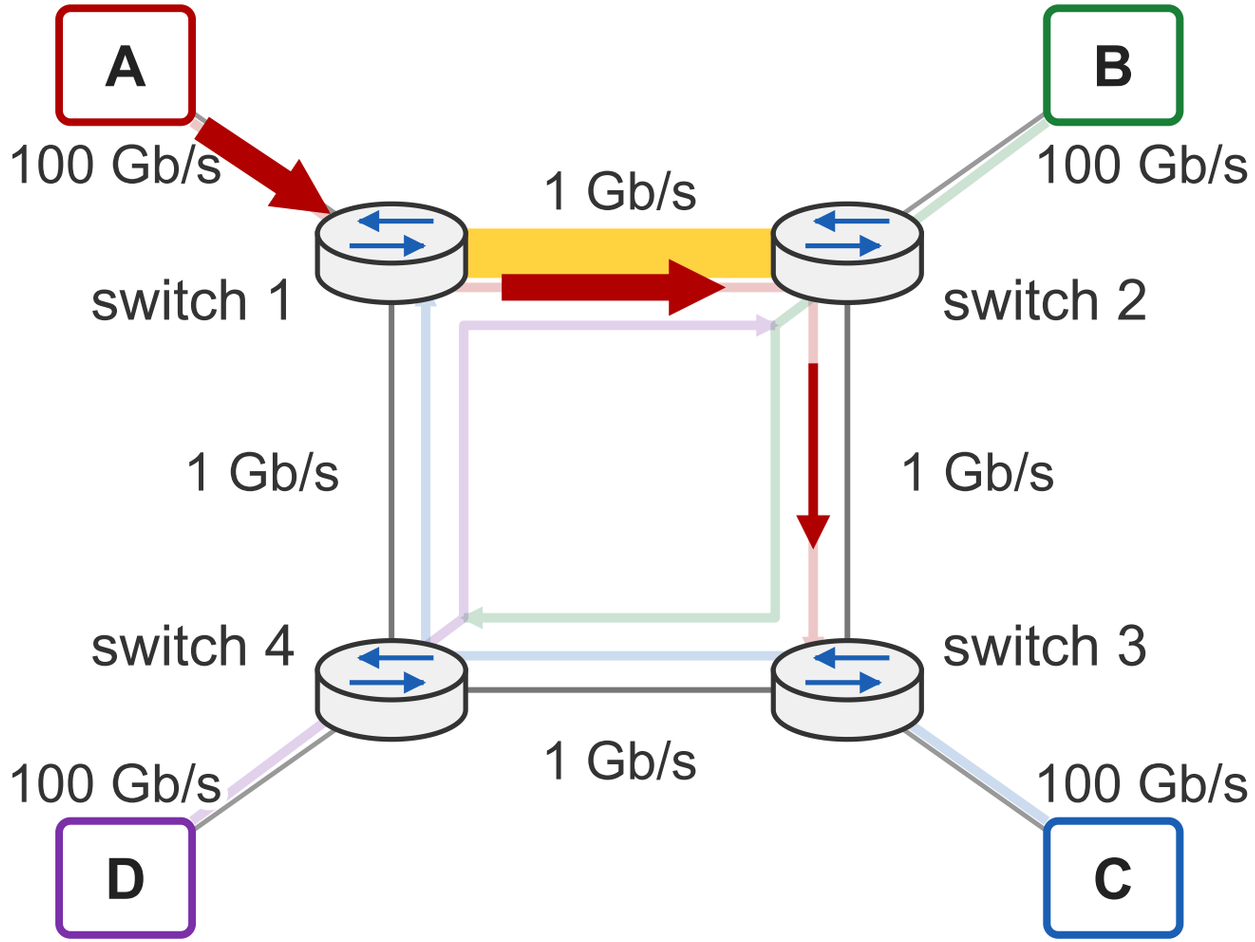


At switch 1, A is in B's place:

- A sends 100 Gb/s
- D's packets at most 1 Gb/s
- A's traffic is getting most of the 1 Gb/s
- So, X is almost 1 Gb/s

Congestion collapse

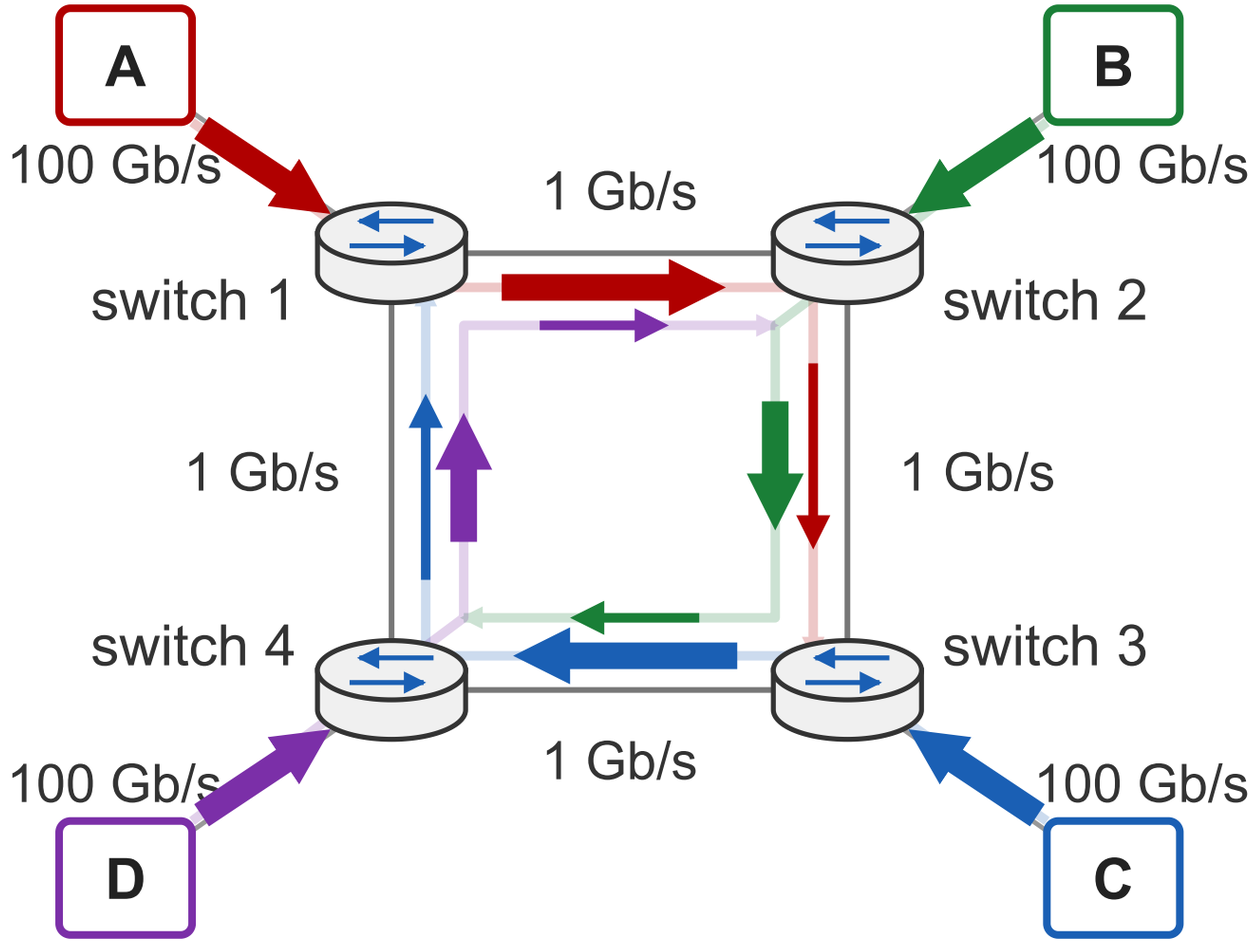
Each host sends to the opposite corner (A → C, B → D, C → A, D → B), as fast as it can.



- So, A is taking most of the link from switch 1 to switch 2
- It lets almost nothing of D's packets through
- But then, almost all of its own packets are dropped at switch 2
  - Because B doesn't let A's packets through, similar to how A didn't let D's packets through

Congestion collapse

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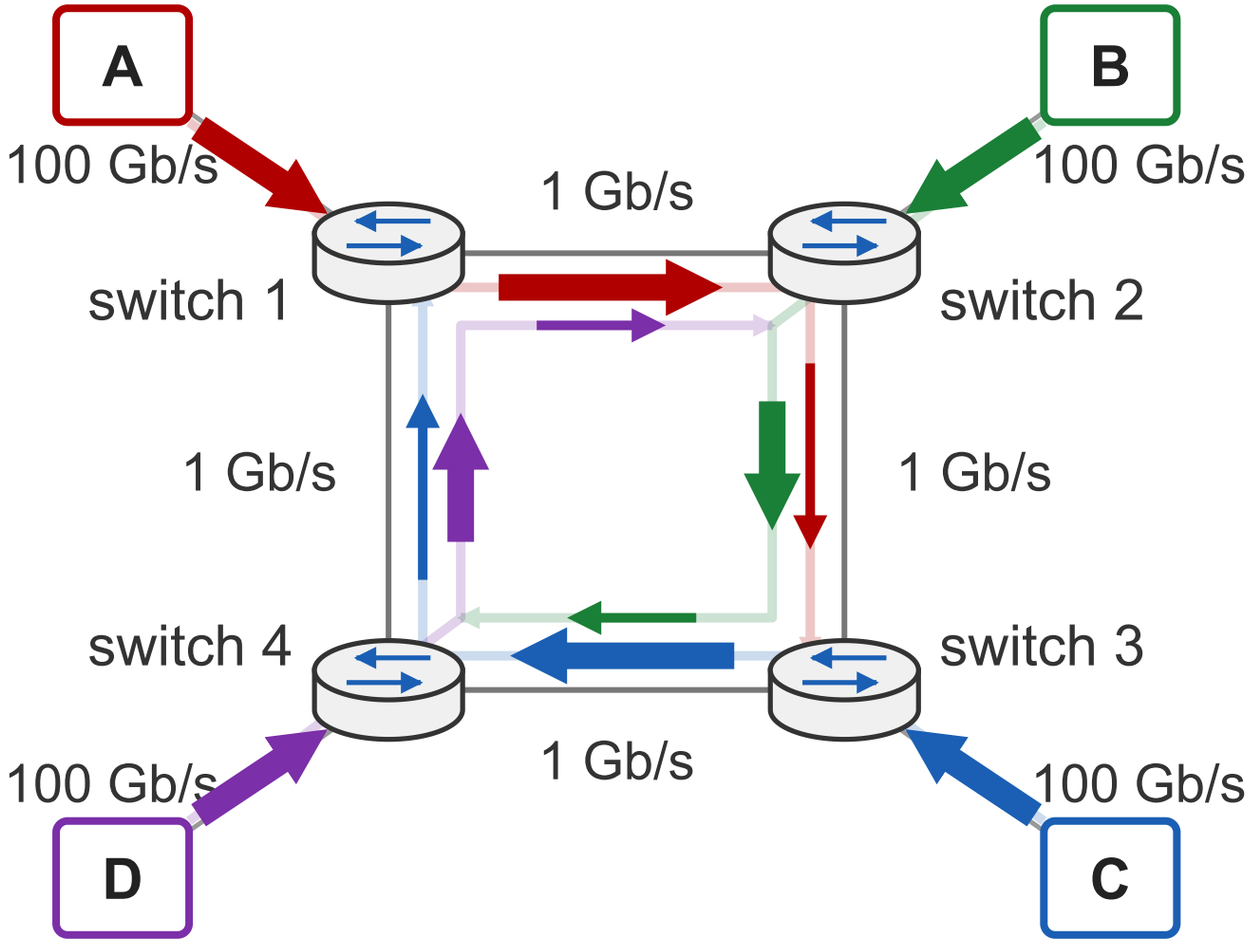


This is the same for every flow:

- Takes most of the bottleneck at its first hop
- Crowds out the other flow on that link
- Then, it is mostly dropped at the next hop

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This is the same for every flow:

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The network links are always busy, but they are not actually delivering data to the end points

How could this have been better? (in-class)

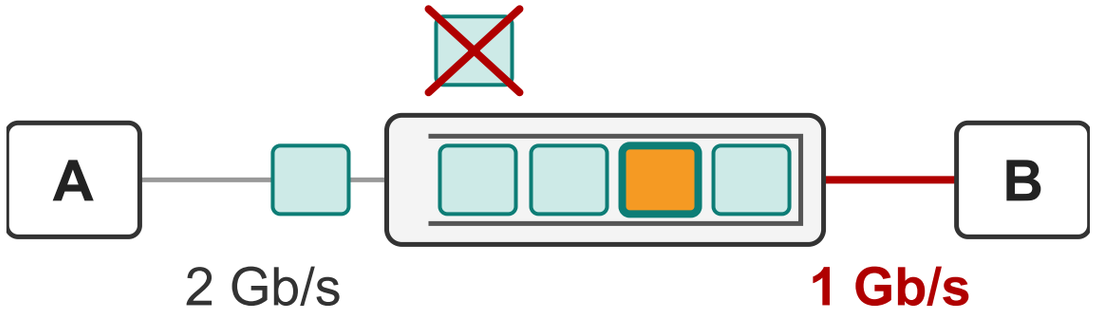
(in-class discussion)

- How much would you have each host send instead?

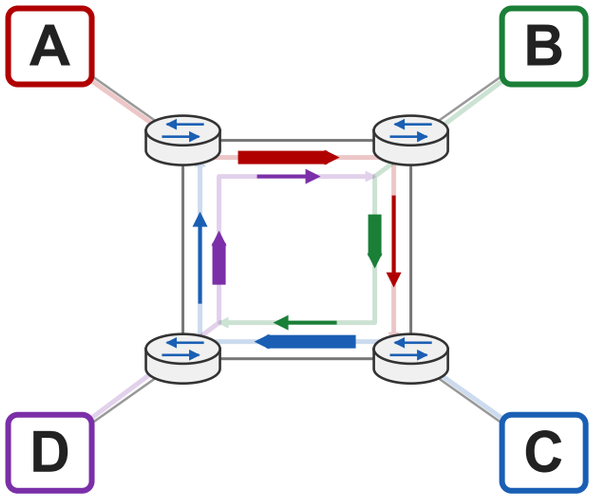
Reset

Recap so far: send only as much as your share of the bottleneck link allows

Only flow on the path: sending faster than the bottleneck link does not get more data through. If anything, it adds losses and wasted sends.

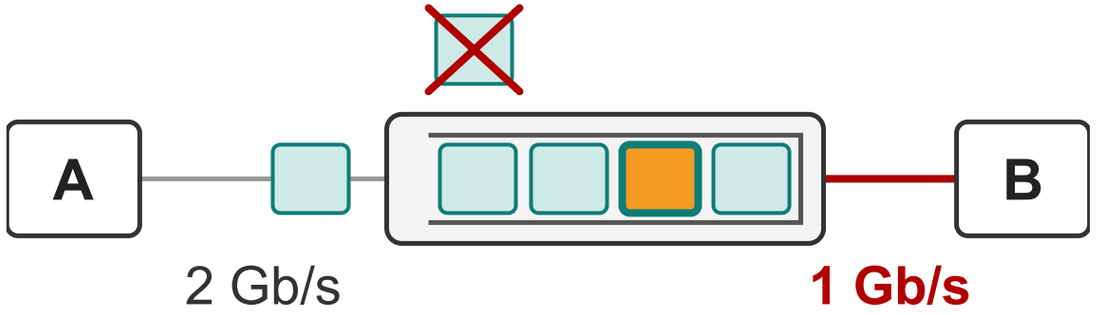


Multiple flows sharing bottleneck links: the network can be busy and still deliver almost nothing (congestion collapse) if flows do not adjust to their share.



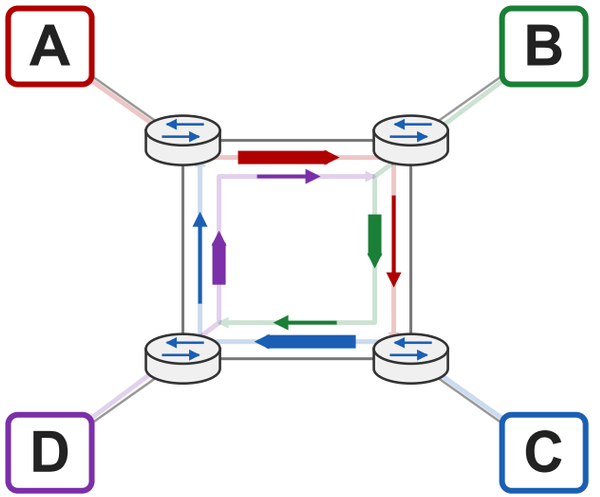
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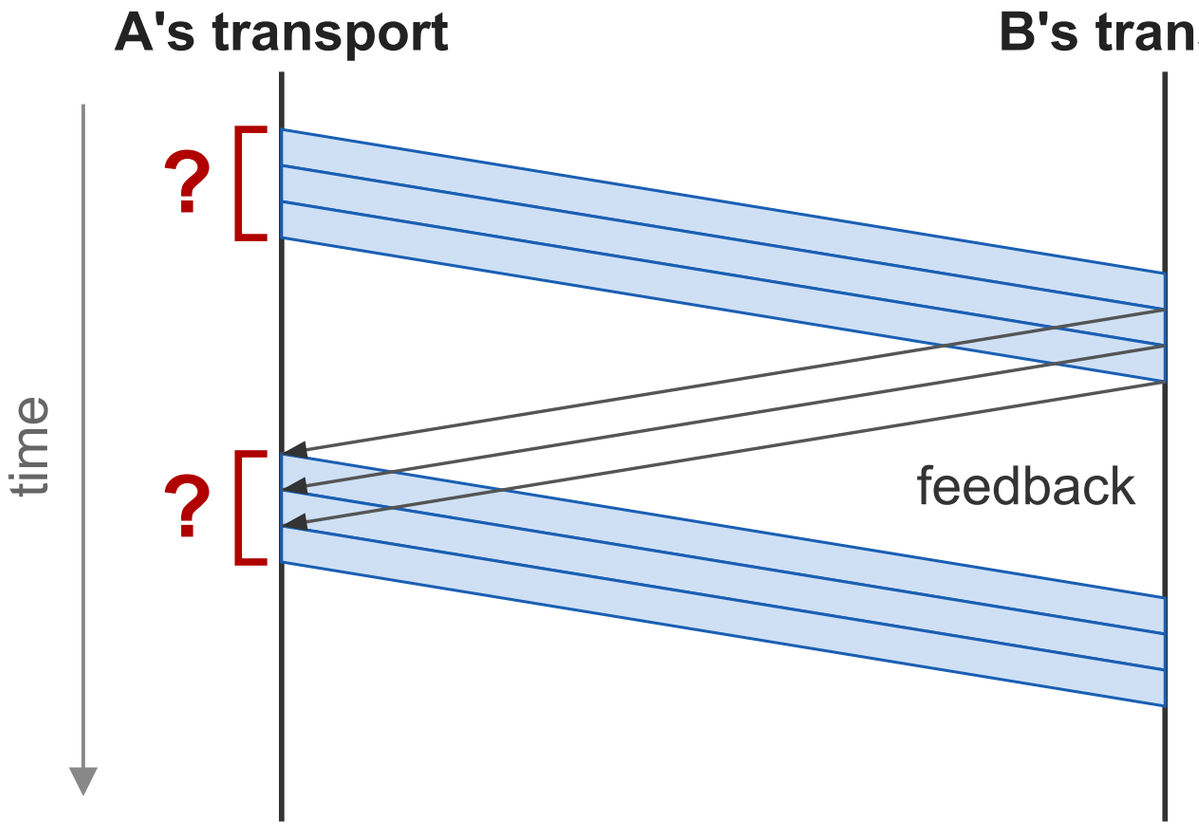


Multiple flows sharing bottleneck links: the network can be busy and still deliver almost nothing (congestion collapse) if flows do not adjust to their share.

One common way: bound the number of in-flight bytes

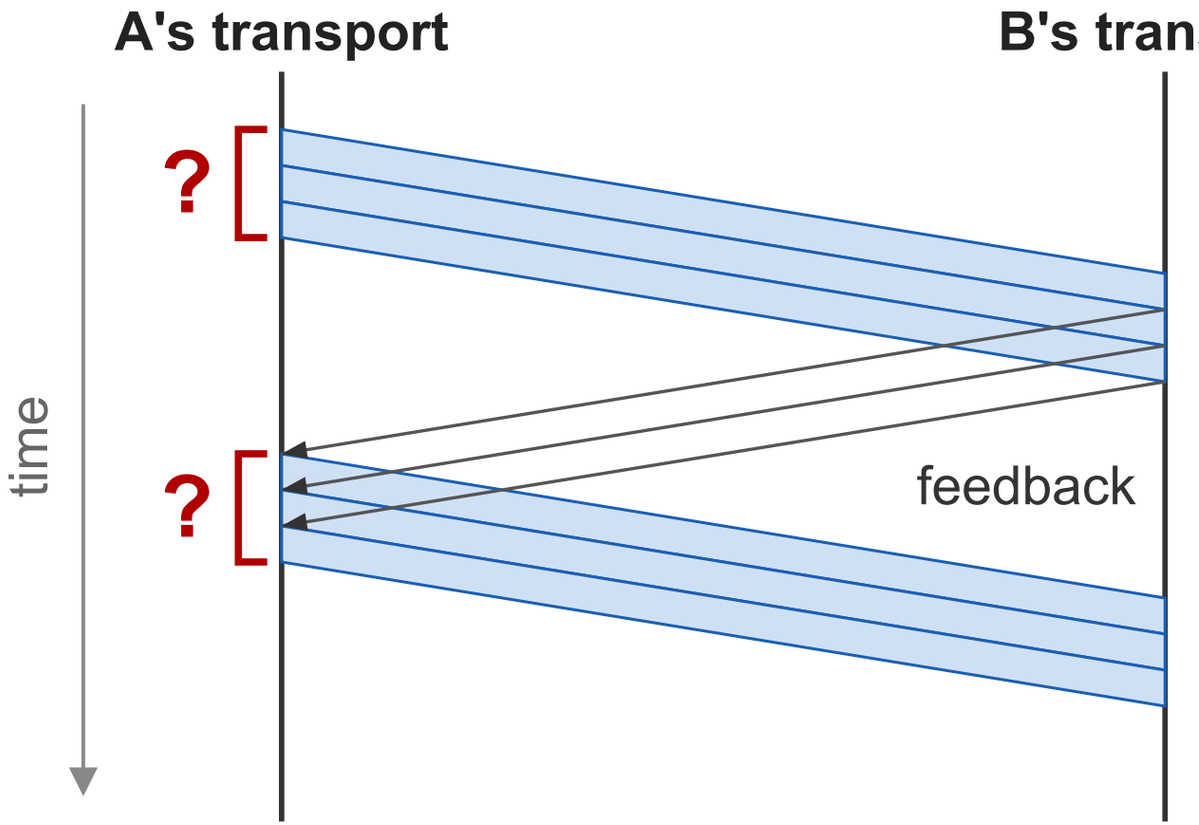


Back to the batch size



A sends a batch back to back, without waiting for feedback in between: one batch every round trip.

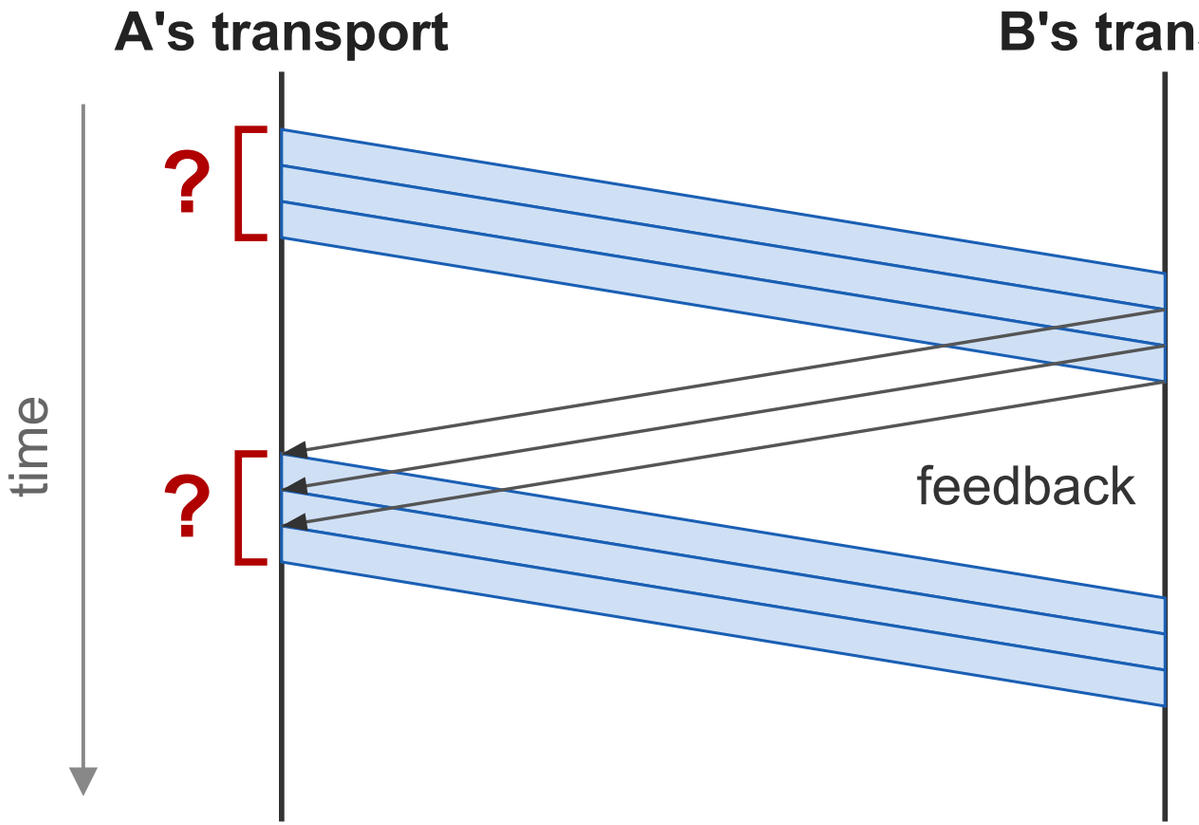
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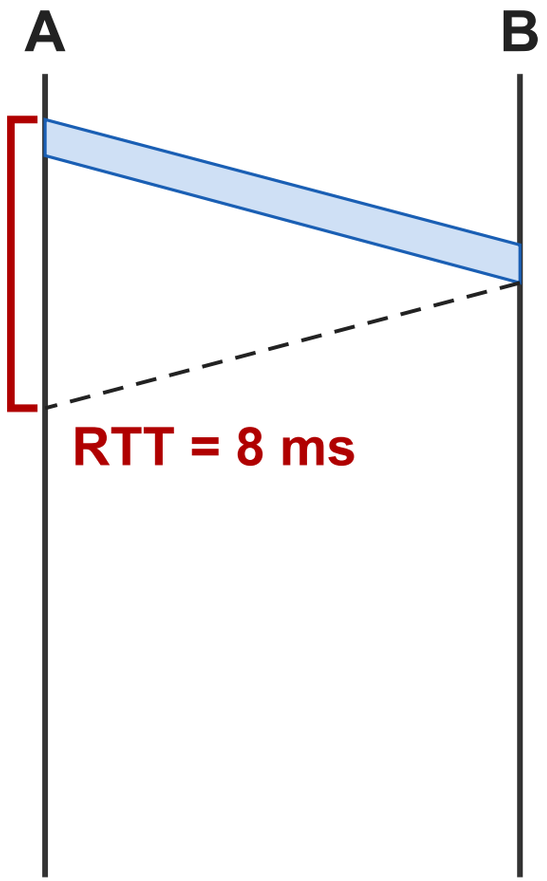


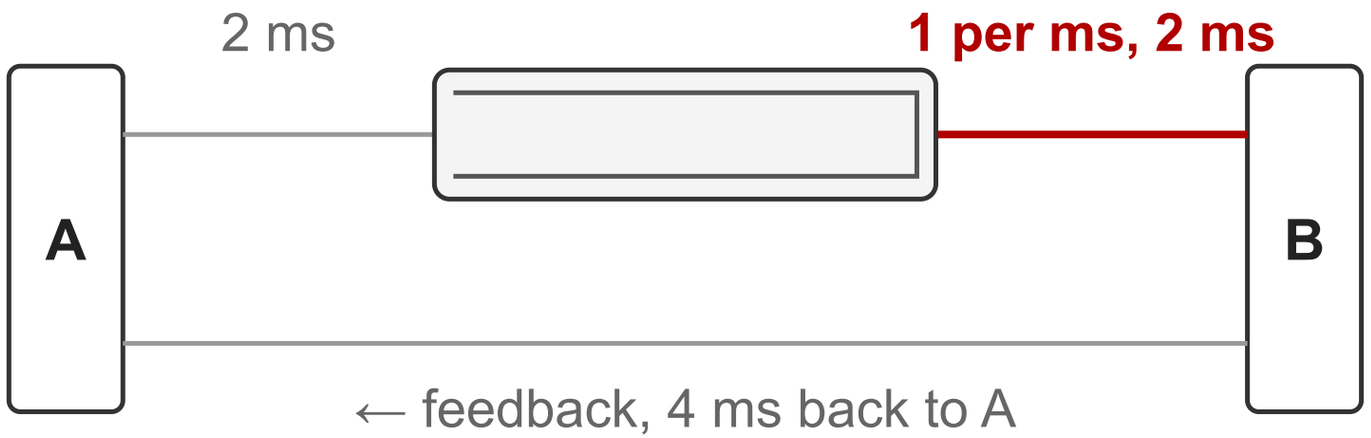
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So what is the right batch size?

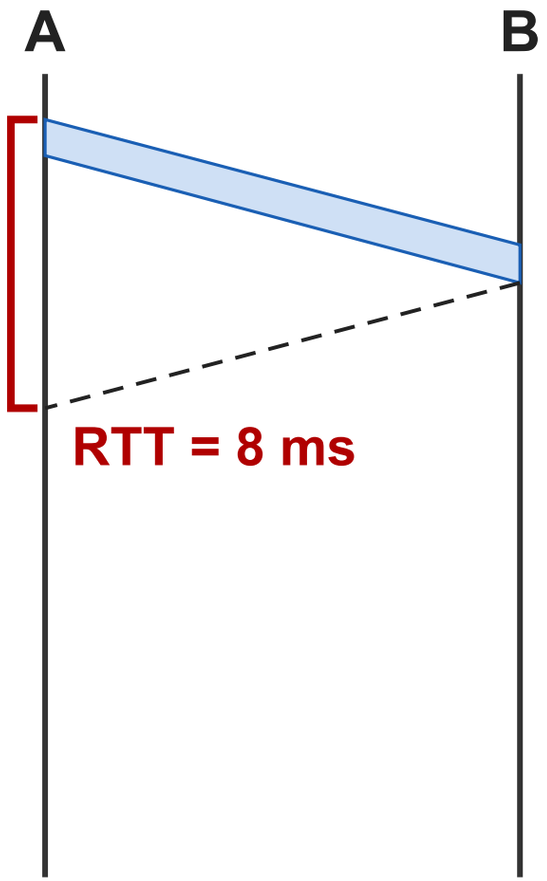
Bounding the number of in-flight bytes

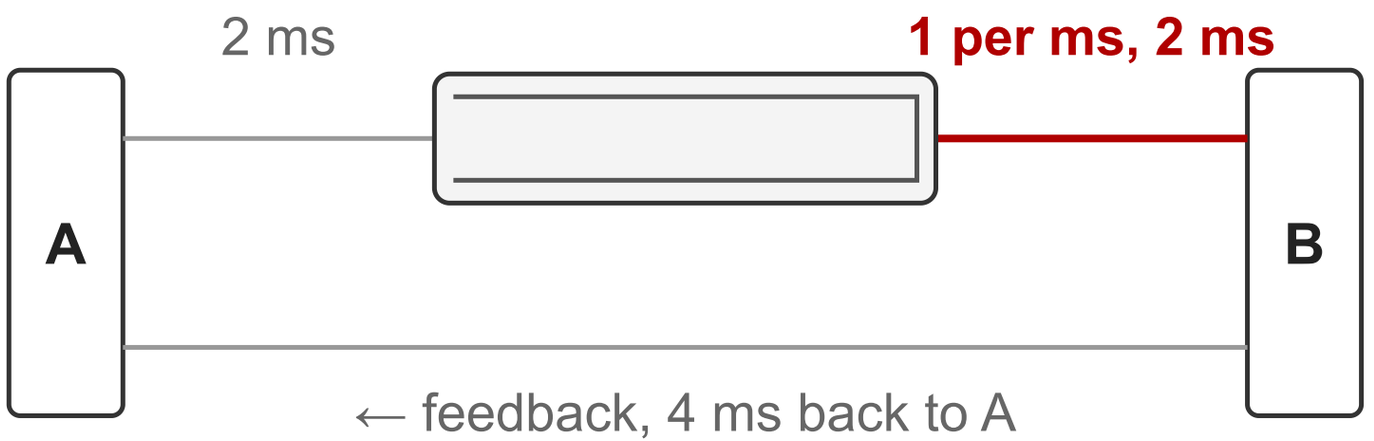




Each packet is 1,500 bytes. One packet and its feedback take one RTT: 8 ms, with nothing waiting in the queue.

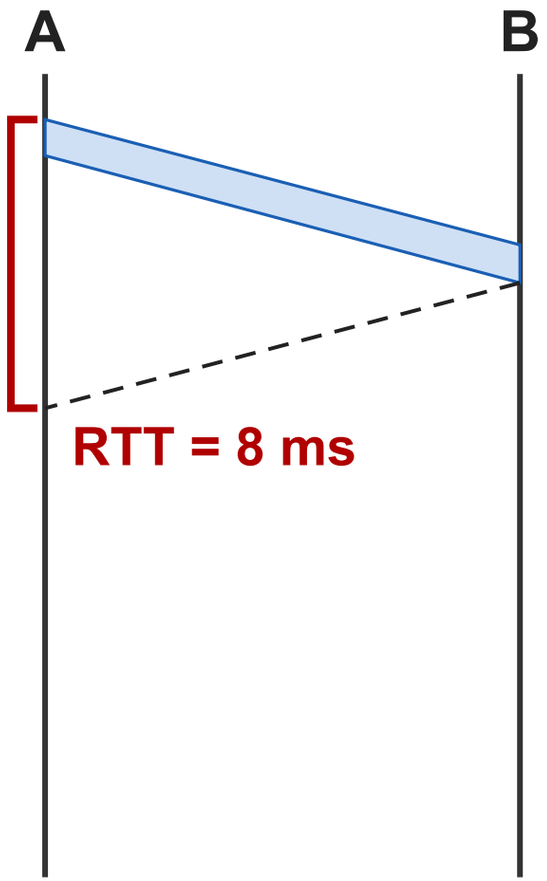
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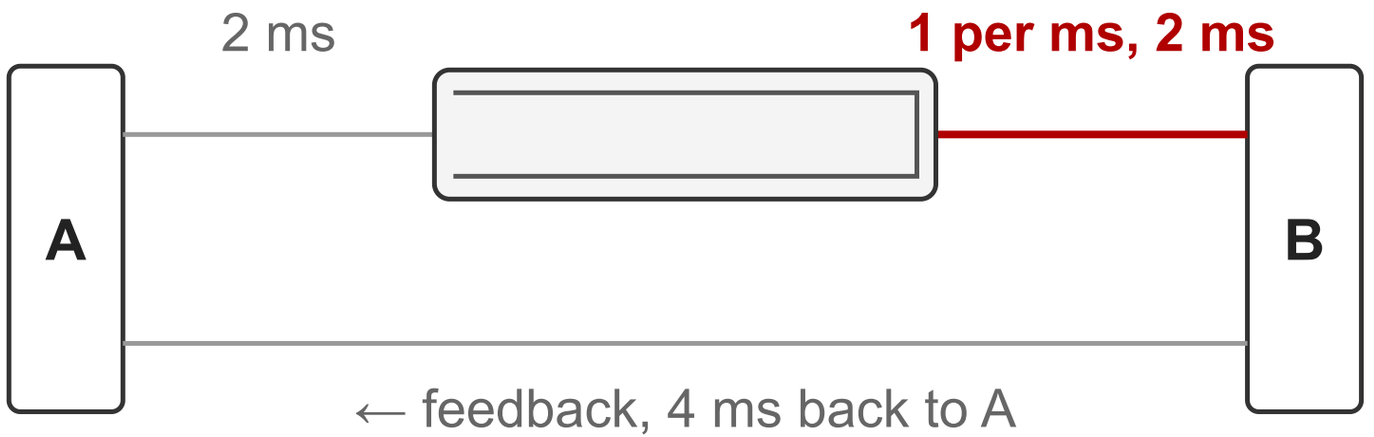




How many bytes could the bottleneck send in one RTT?

Bounding the number of in-flight bytes

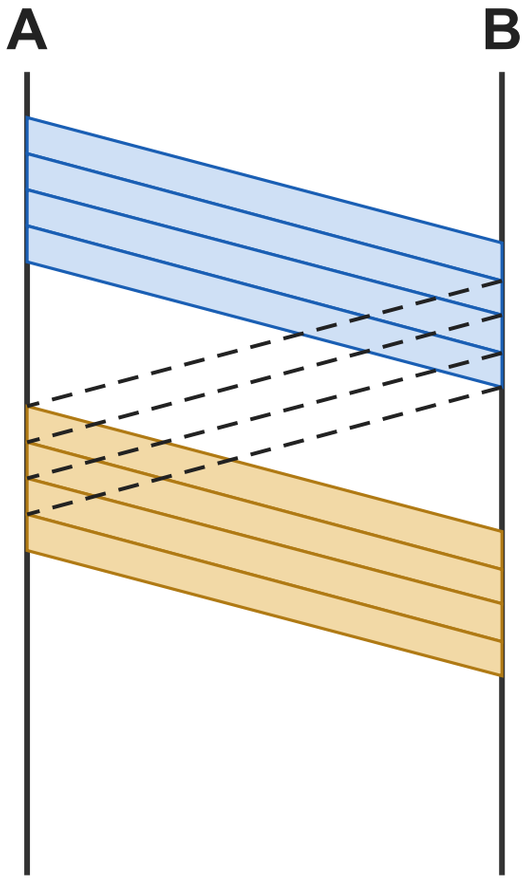


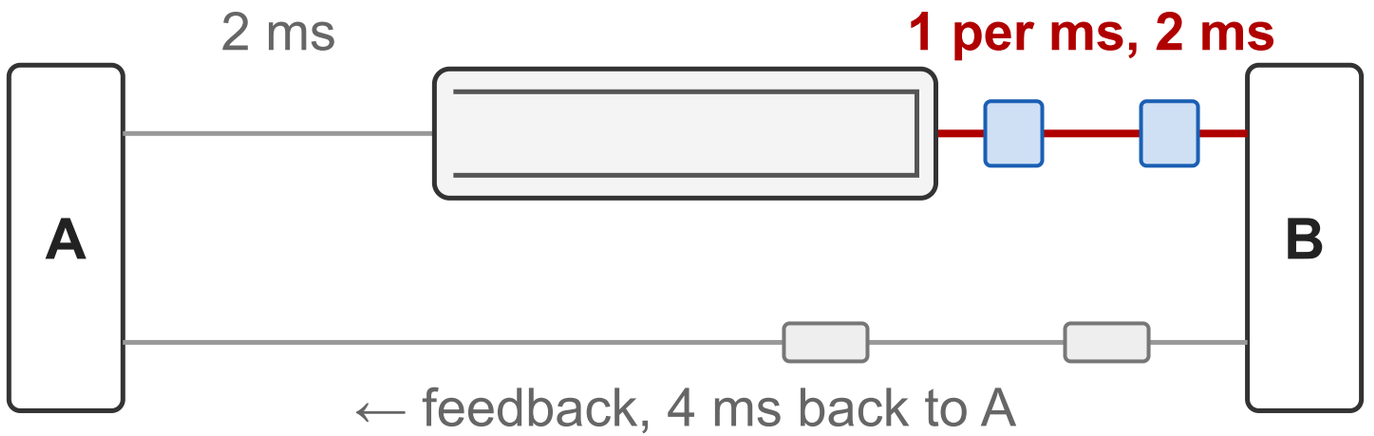


How many bytes could the bottleneck send in one RTT?

1 packet per ms × 8 ms = 8 packets = 12,000 bytes: the bandwidth-delay product (BDP)

Bounding the number of in-flight bytes

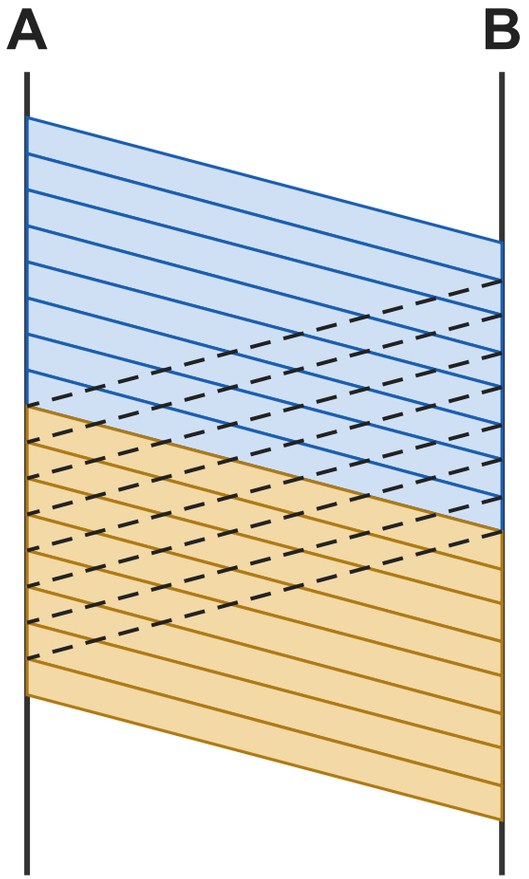


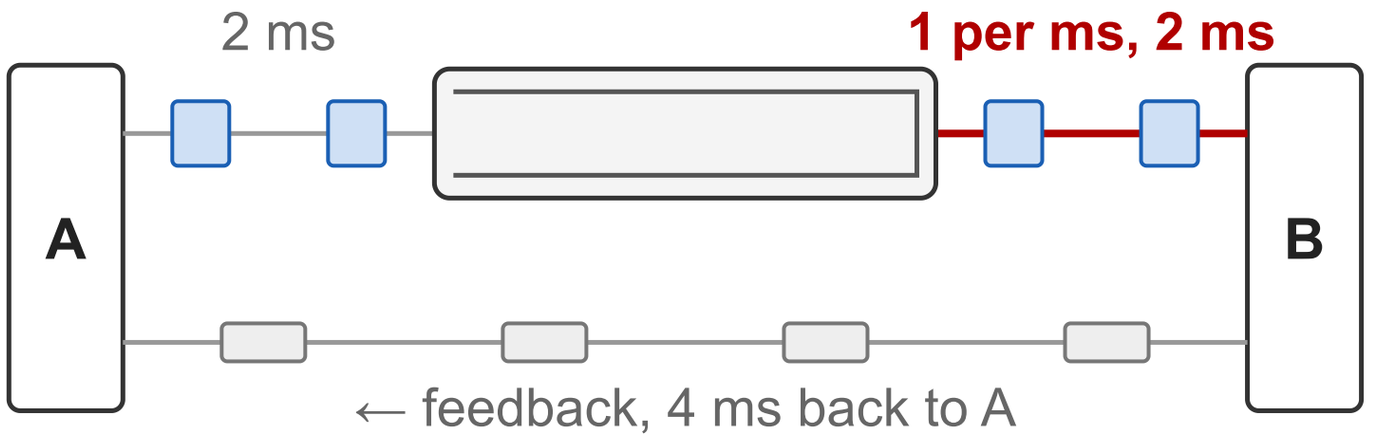


Bound 6,000 bytes (4 packets): below the BDP

A sends 4 back to back, then waits for the first feedback. B receives 4 packets, then sits idle for 4 ms.

Bounding the number of in-flight bytes

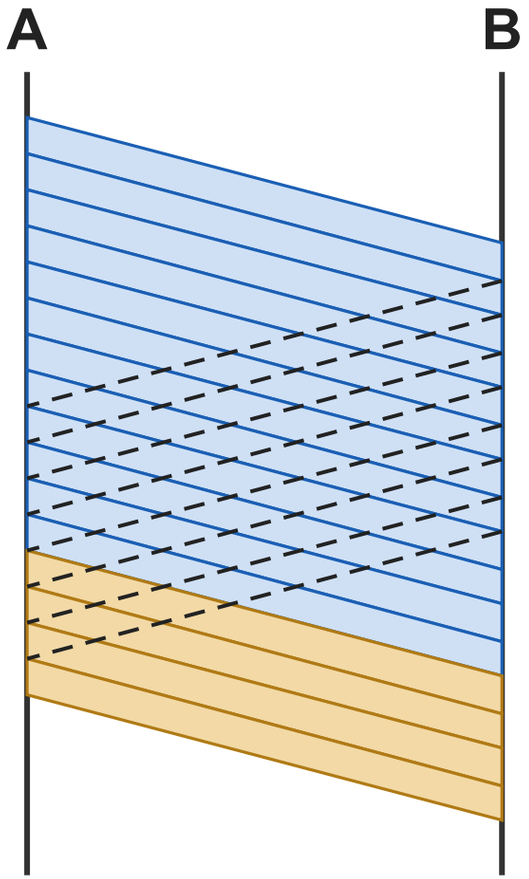


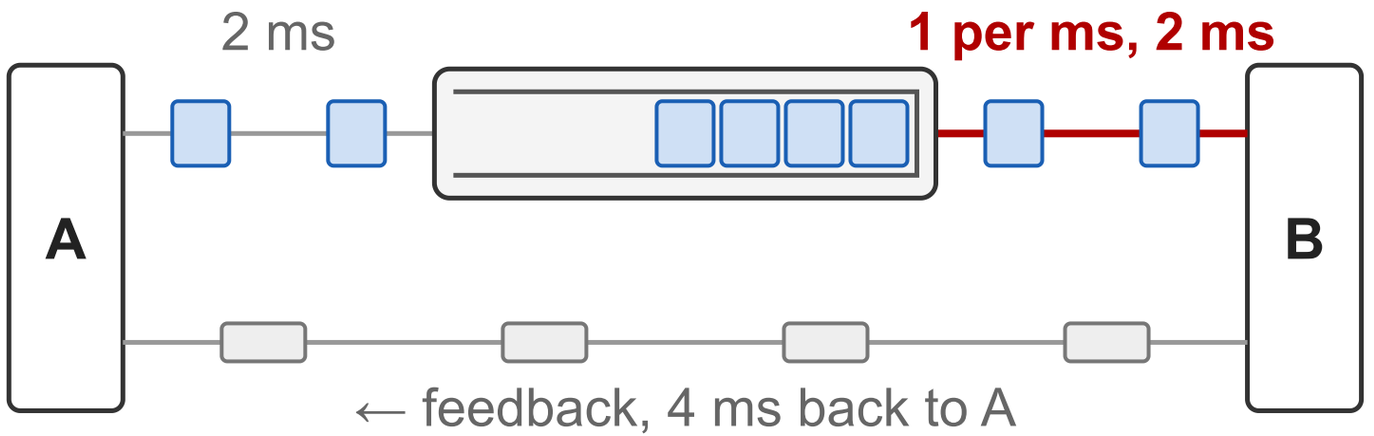


Bound 12,000 bytes (8 packets): the BDP

The round trip is full, the queue empty.
B continuously receives packets, and
there is no idle time.

Bounding the number of in-flight bytes

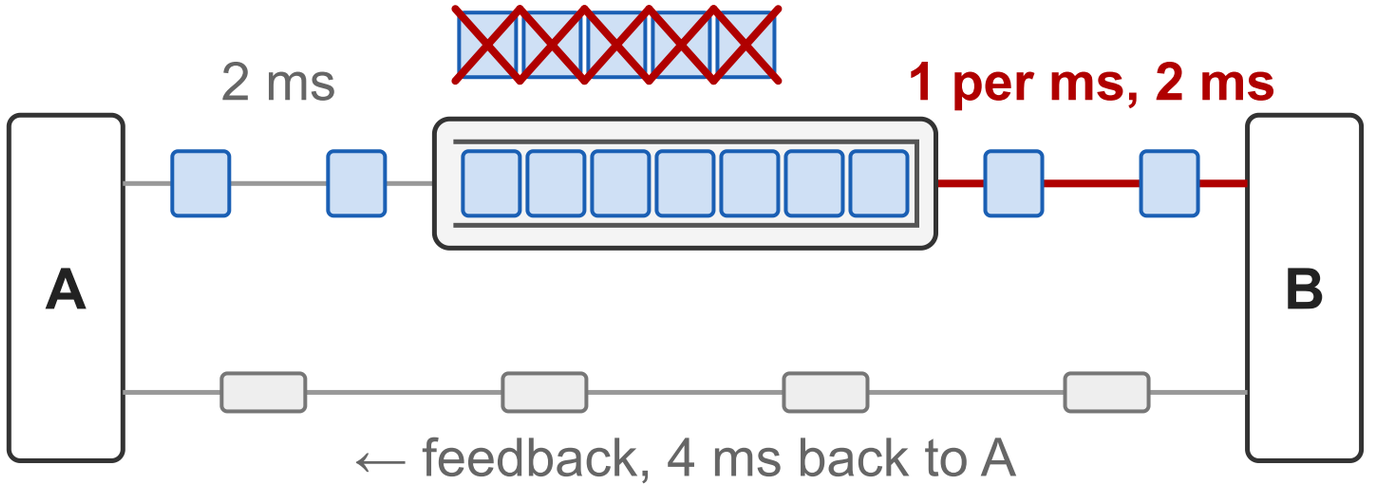




Bound 18,000 bytes (12 packets): above the BDP

The round trip is full **and** four packets always wait in the queue. B still gets one every 1 ms, with no idle time. But the round-trip time is now 12 ms.

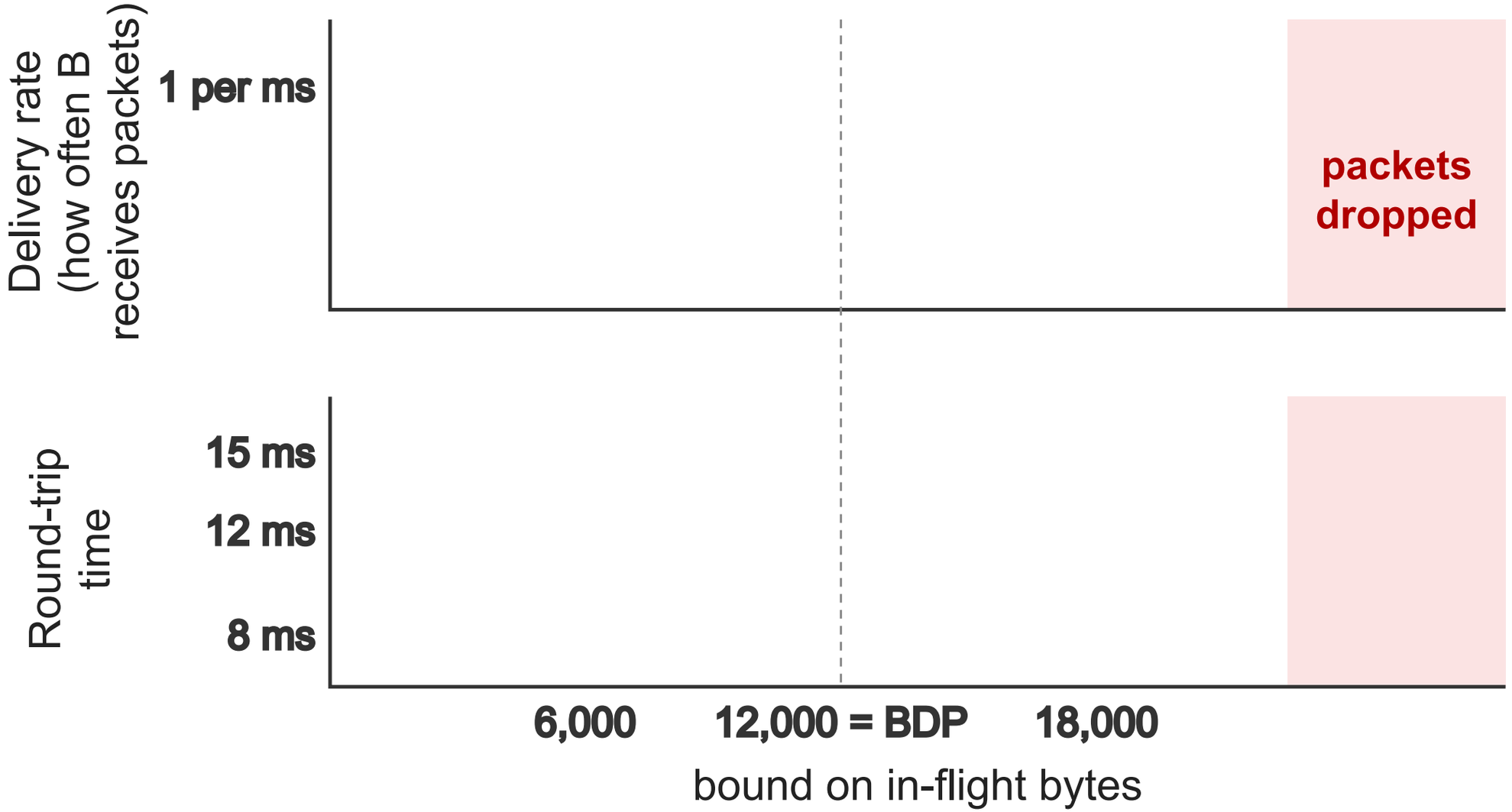
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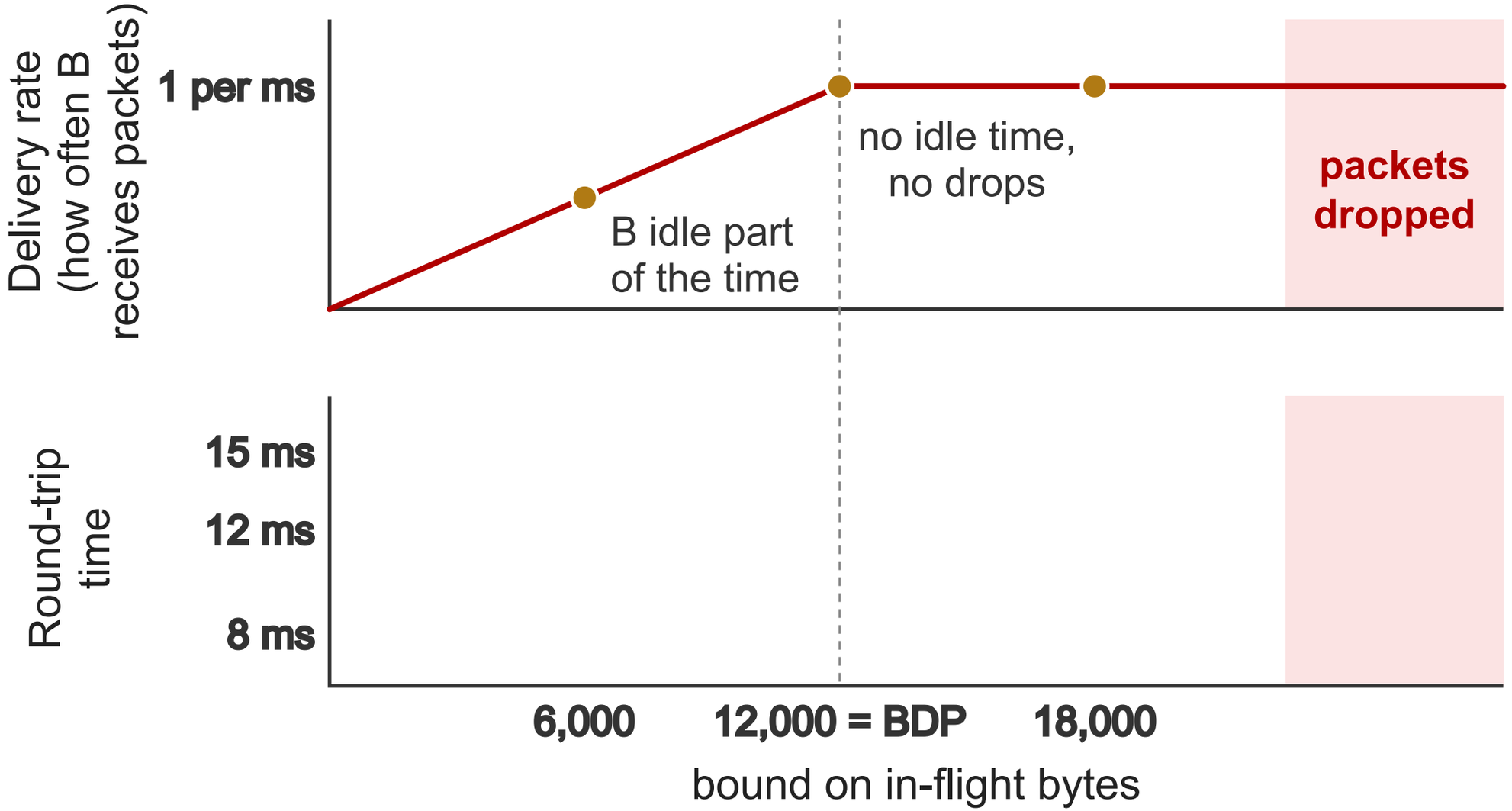
Bound 30,000 bytes (20 packets): far above the BDP

Round trip full, queue full (7), 5 dropped every round trip. B still gets one every 1 ms, but some are retransmissions. The round-trip time is now 15 ms.

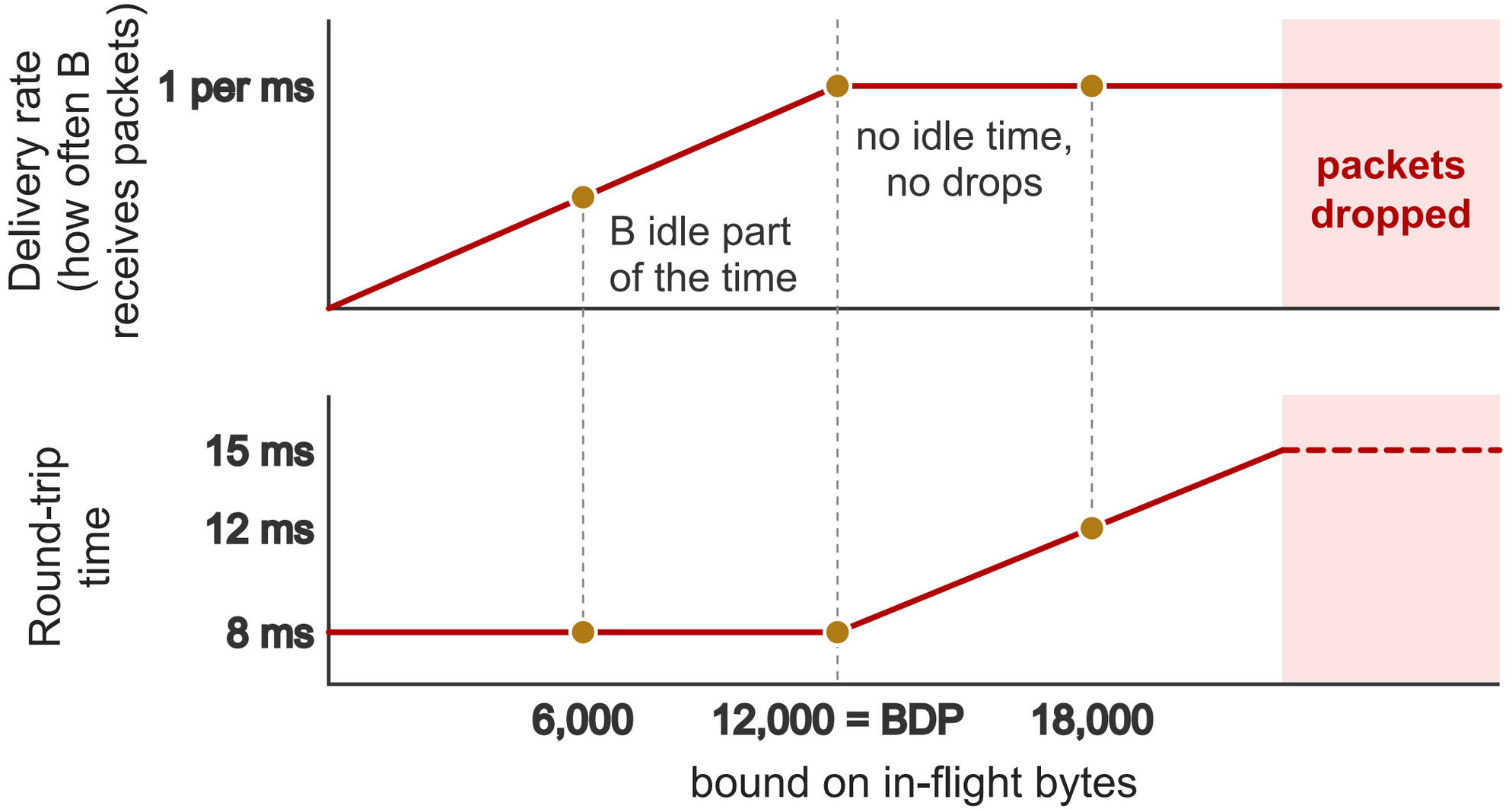
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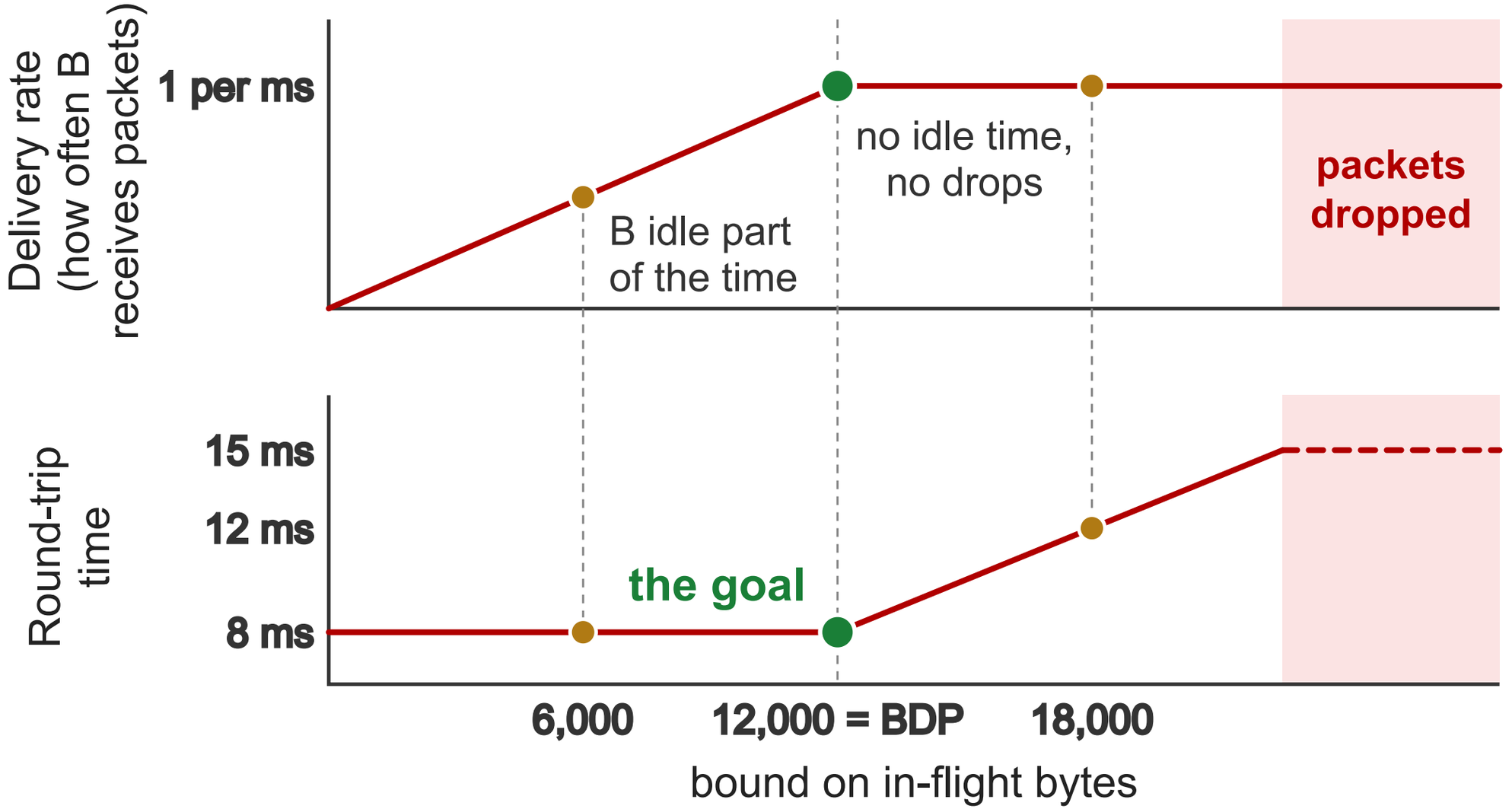
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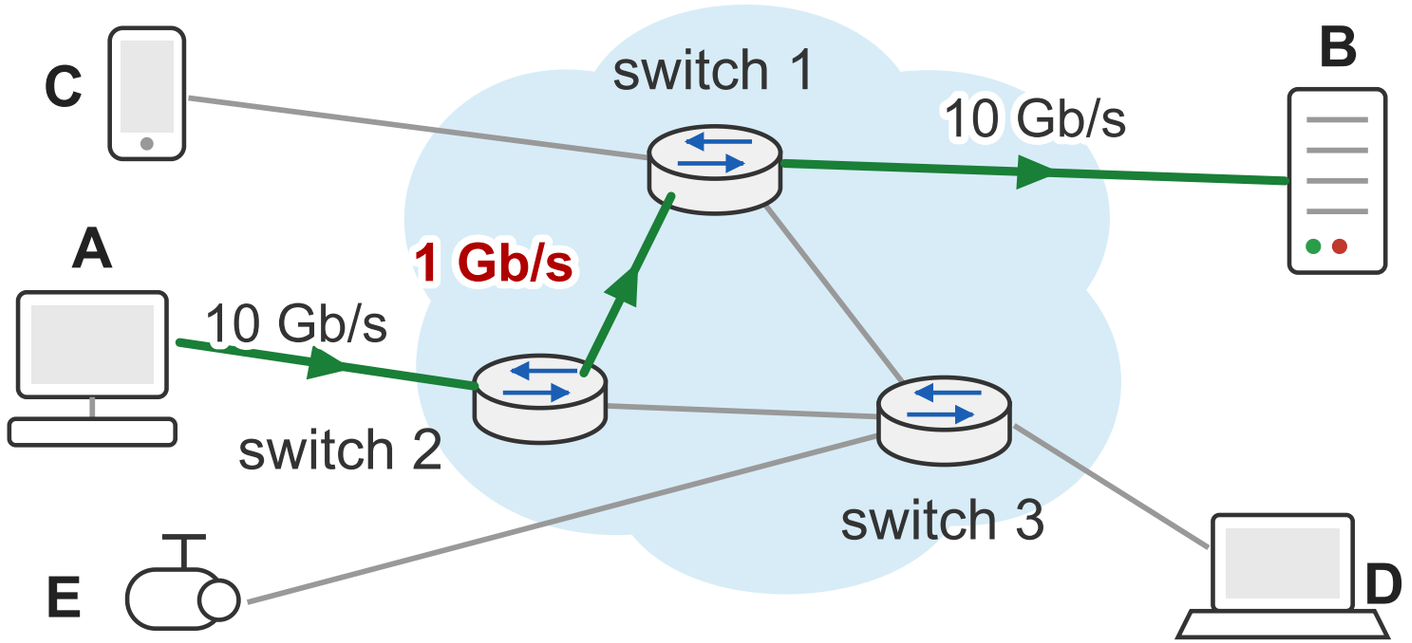
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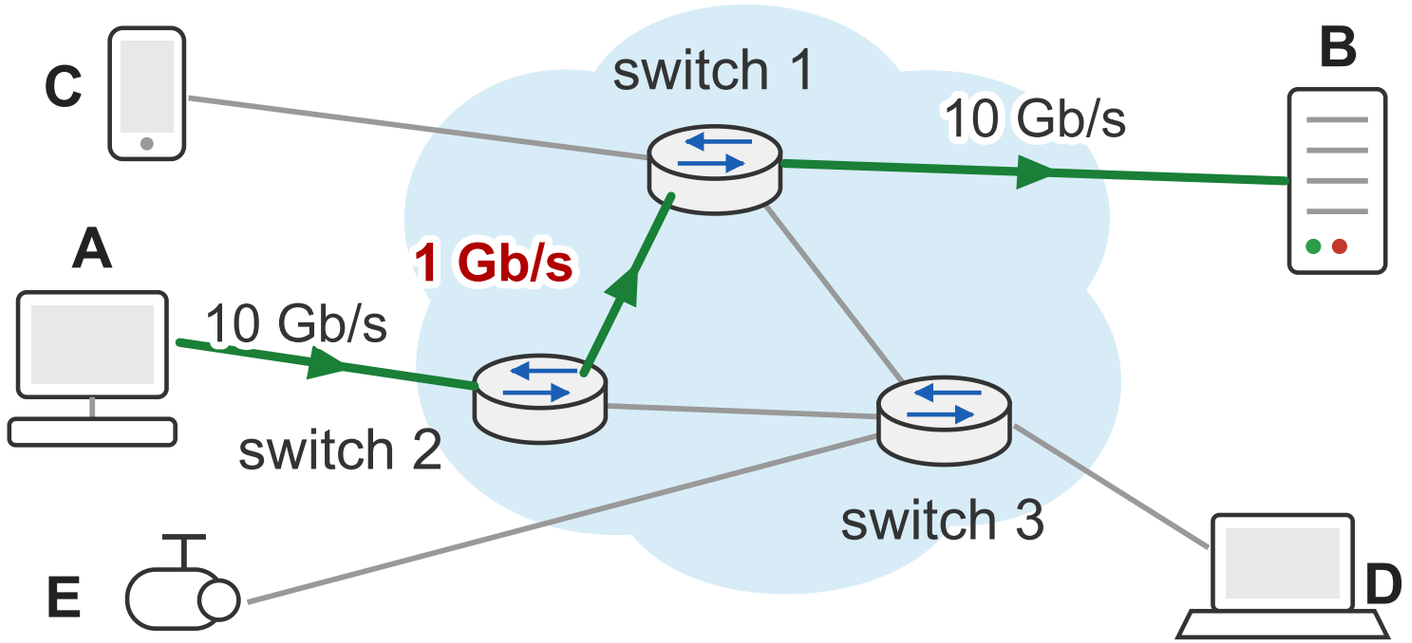
How fast should the sender send?



Consider a sender and a receiver:

- **The sender knows exactly what path its traffic will take to the destination**, and the properties (rate, propagation delay, etc.) of each link
- The sender always has more data to send

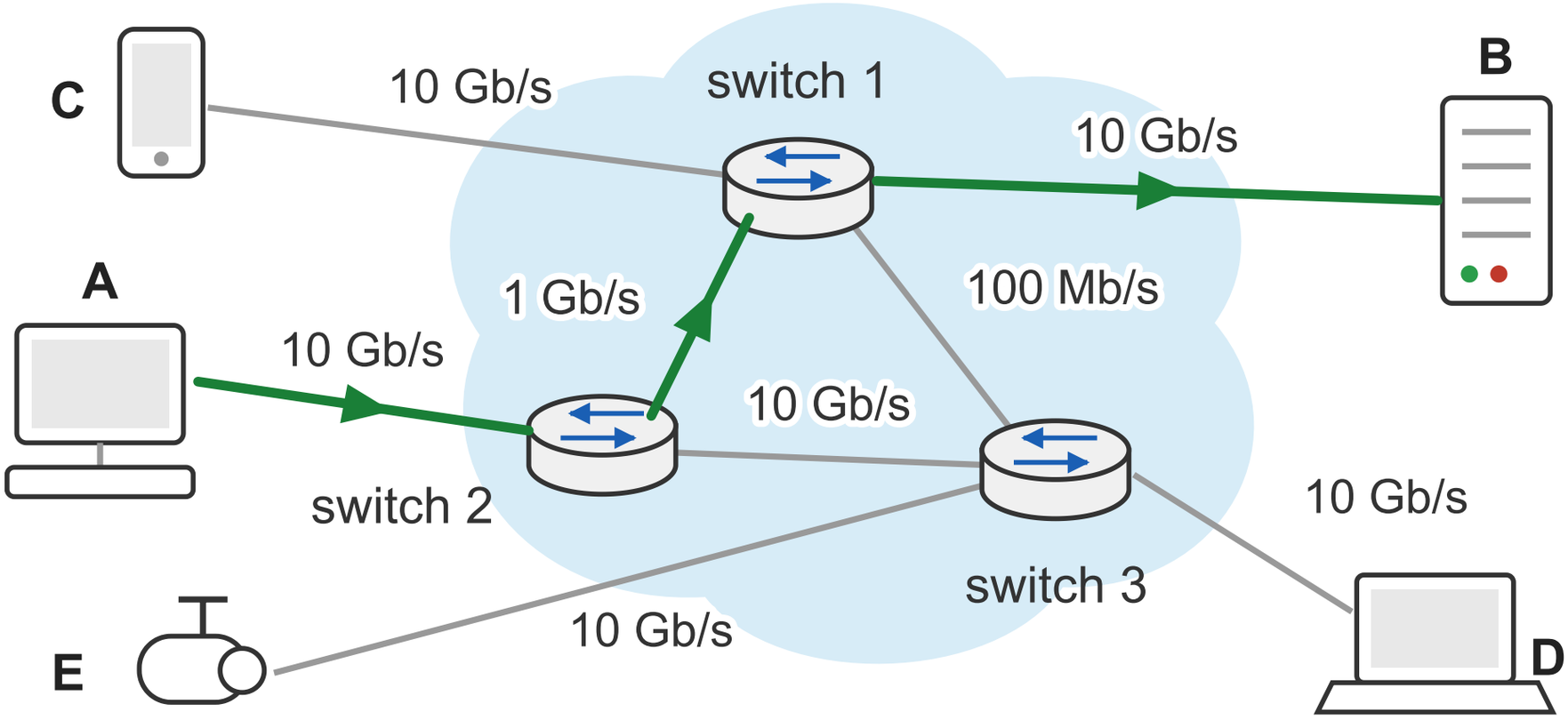
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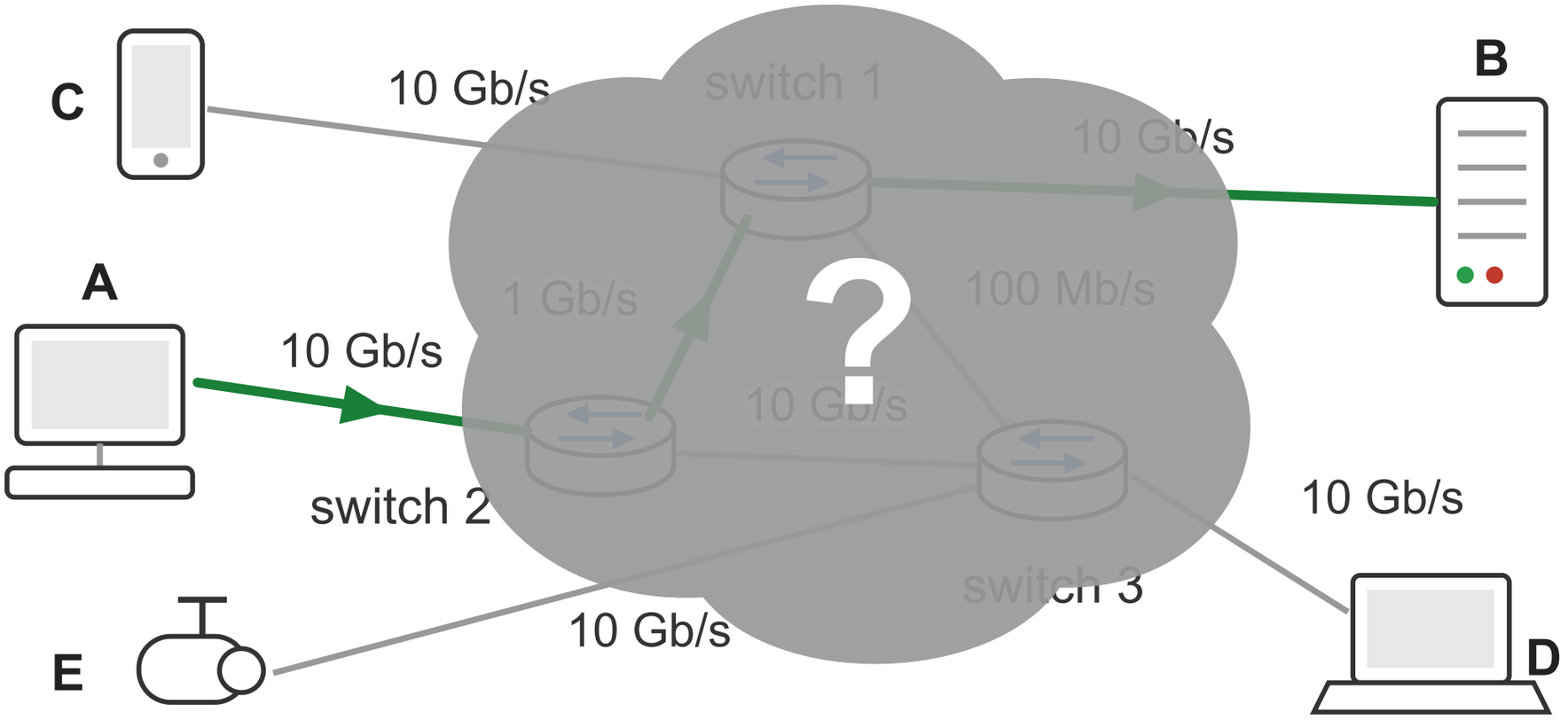
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In reality



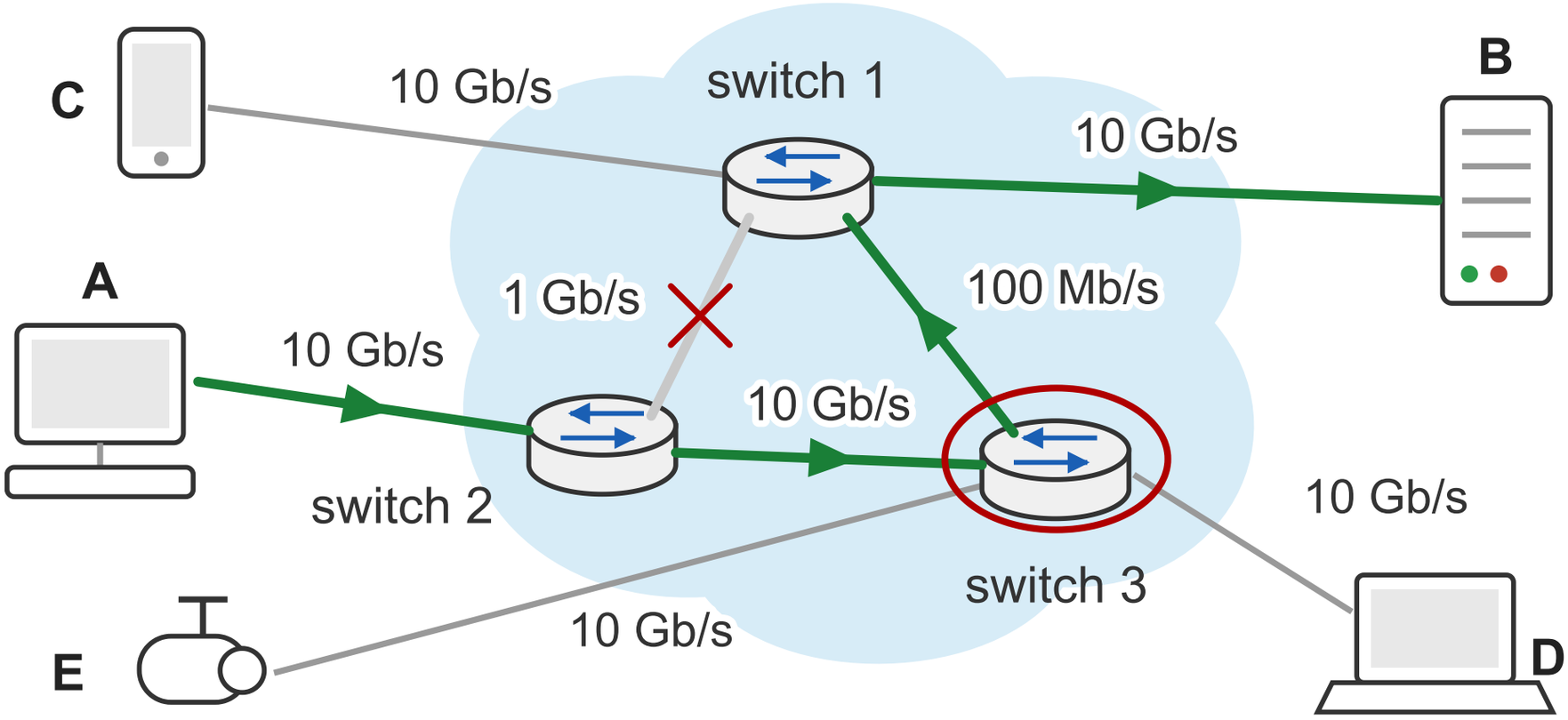
So far: A knew its path (green) and every link's rate

In reality



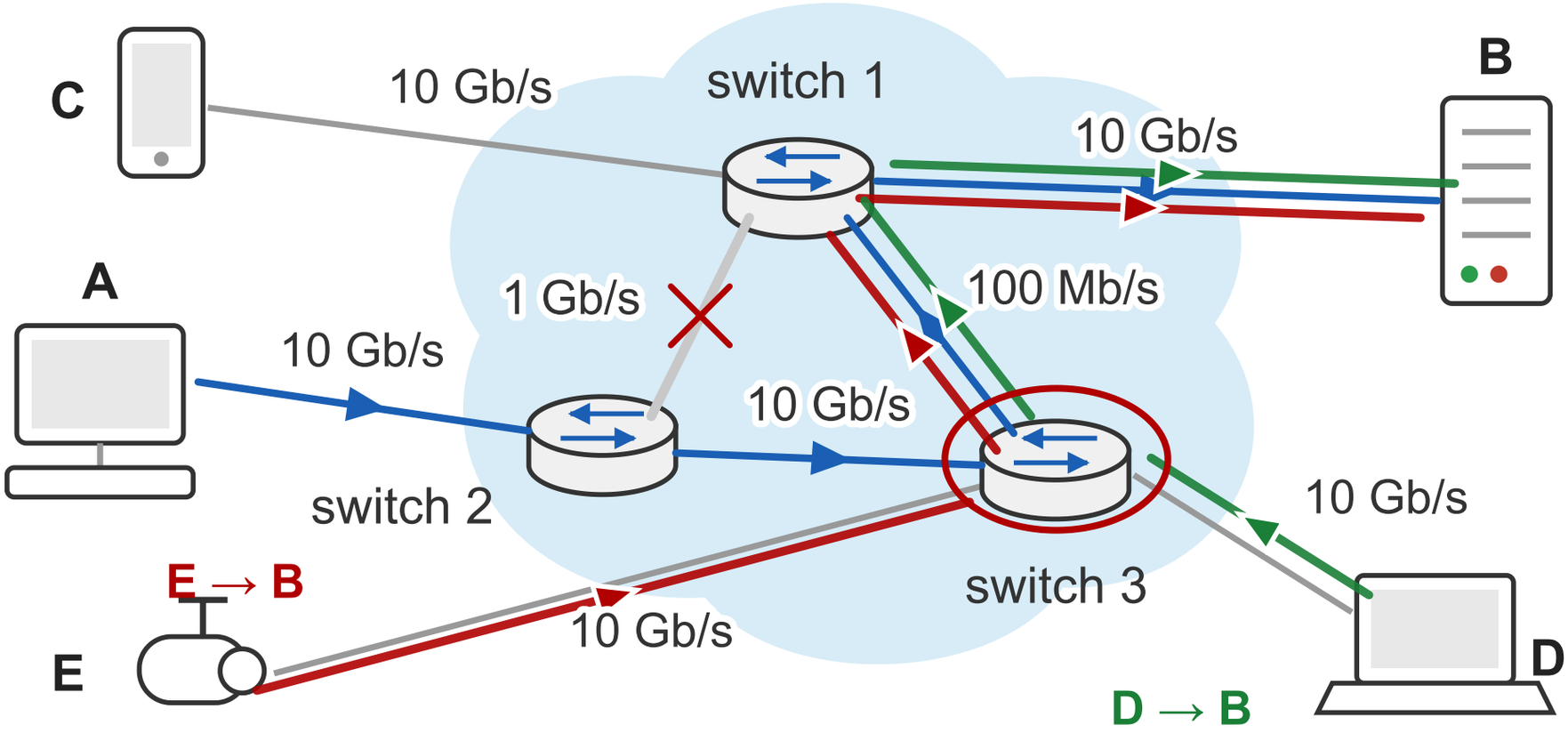
1. A does not know its path, or the rates on it

In reality



2. Even if A somehow found out once, that path might change (link failures, traffic changes, …)

In reality



3. Other flows may start sharing the bottleneck, and reduce A's share

How would you find the right share? (in-class)

(in-class discussion)

- You are a sender communicating with a receiver. You don’t get any extra information from the network or other senders and receivers that share the network with you.
- How can you figure out your “right” share, and notice when it changes?

Reset

Congestion control

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  - Share fairly with other flows

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  - …
- We'll see example solutions in the next lecture.